A subclass of exotic symbols

Zipeng Wang

Abstract

We investigate a new subclass of Hörmander exotic symbols: $\mathbf{S}^0_{\rho\ \delta}$ which satisfies a differential inequality with a mixture of homogeneities. By taking singular integral realization, the regarding kernels have certain characteristic properties on product spaces. We prove that pseudo-differential operators with symbols belonging to this class are bounded on $\mathbf{L}^p(\mathbb{R}^{\mathbf{N}})$ for $1 < p < \infty$. Moreover, they form an algebra.

1 Introduction

Let $0 < \delta \leq \rho < 1$. A symbol $\sigma(x, \xi) \in C^\infty(\mathbb{R}^{\mathbf{N}} \times \mathbb{R}^{\mathbf{N}})$ belongs to the exotic symbol class: $\mathbf{S}^m_{\rho\ \delta}$ if

$$\left|\partial_\xi^\alpha \partial_x^\beta \sigma(x, \xi)\right| \ \leq\ \mathfrak{B}_{\alpha\ \beta}\ (1 + |\xi|)^{m-\rho|\alpha|+\delta|\beta|} \tag{1. 1}$$

for every multi-indices α, β.

$\diamond$ Throughout, $\mathfrak{B} > 0$ is regarded as a generic constant depending on the subindices.

Consider $\mathbf{R}$ as a distribution in $\mathbb{R}^{\mathbf{N}}$ agree with

$$\mathbf{R}(x) \ =\ e^{\mathbf{i}/|x|} \left(\frac{1}{|x|}\right)^{\frac{3}{2}} \phi(|x|), \qquad x \neq 0$$

where ϕ is a smooth cut-off function vanish outside the unit ball. The Fourier transform of $\mathbf{R}$ belongs to $\mathbf{S}^0_{\frac{1}{2}\ 0}$. This distribution is known as *Riemann singularity*. Convolutions with $\mathbf{R}$ are bounded on $\mathbf{L}^2(\mathbb{R}^{\mathbf{N}})$ only. More details can be found in Chapter VII of Stein [7].

In this paper, we introduce a new class of symbols. Let $\mathbb{R}^{\mathbf{N}} = \mathbb{R}^{\mathbf{N}_1} \times \mathbb{R}^{\mathbf{N}_2} \times \cdots \times \mathbb{R}^{\mathbf{N}_n}, n \geq 2$.

Symbol class $\mathbf{S}^0_\rho$: We say $\sigma \in \mathbf{S}^0_\rho, 0 < \rho < 1$ if

$$\left|\partial_\xi^\alpha \partial_x^\beta \sigma(x, \xi)\right| \ \leq\ \mathfrak{B}_{\alpha\ \beta}\ \prod_{i=1}^n \left(\frac{1}{1 + |\xi_i| + |\xi|^\rho}\right)^{|\alpha_i|} (1 + |\xi|)^{\rho|\beta|} \tag{1. 2}$$

for every multi-indices α, β.

The original version of $\mathbf{S}^0_\rho$ for $\rho = \frac{1}{2}$ is first suggested by Stein. This symbol class arises by considering compositions of Calderón-Zygmund operators with different homogeneities of given dilations, i.e: one isotropic and the other parabolic. A systematic discussion for such singular integral operators having a multi-parameter characteristic has been established in the Memoirs paper by Nagel, Ricci, Stein and Wainger [1].

Observe that $\mathbf{S}^0_\rho \subset \mathbf{S}^0_{\rho,\rho}$. We assert

$$\mathbf{T}f(x) \;=\; \int_{\mathbb{R}^\mathbf{N}} e^{2\pi \mathbf{i} x\cdot\xi}\sigma(x,\xi)\widehat{f}(\xi)d\xi, \qquad \sigma\in\mathbf{S}^0_\rho. \tag{1. 3}$$

The study of certain operators which commute with a multi-parameter family of dilations, dates back to the time of Jessen, Marcinkiewicz and Zygmund. Over the several past decades, a number of pioneering results have been accomplished, for example by Fefferman [9]-[10], Fefferman and Stein [6], Cordoba and Fefferman [11], Chang and Fefferman [12], Journé [13], Pipher [14], Carbery and Seeger [17]-[18], Nagel and Stein [8] and Müller, Ricci and Stein [3].

The kernel of $\mathbf{T}$ in (1. 3), denoted by Ω, is a distribution in $\mathbb{R}^\mathbf{N}$ agree with

$$\Omega(x, x-y) \;=\; \int_{\mathbb{R}^\mathbf{N}} e^{2\pi\mathbf{i}(x-y)\cdot\xi}\sigma(x,\xi)d\xi, \qquad x\neq y. \tag{1. 4}$$

Remark 1.1 *Write $z = x-y$. We find $\sigma\in\mathbf{S}^0_\rho, 0<\rho<1$ equivalent to the size estimate and cancellation property in below.*

Size estimate *For every multi-indices α, β, we have*

$$\left|\partial_z^\alpha\partial_x^\beta\Omega(x,z)\right| \;\leq\; \mathfrak{B}_{\alpha\,\beta}\left(\frac{1}{|z|}\right)^{\rho|\beta|}\prod_{i=1}^n\left(\frac{1}{|z_i|+|z|^{1/\rho}}\right)^{\mathbf{N}_i+|\alpha_i|}, \qquad |z|>0. \tag{1. 5}$$

On the other hand, $\partial_z^\alpha\partial_x^\beta\Omega(x,z)$ decays rapidly as $|z|\longrightarrow\infty$.

Let $\varphi_i\in C_o^\infty(\mathbb{R}^{\mathbf{N}_i}), i=1,2,\ldots,n$ be normalized cut-off functions.

Cancellation property *We have*

$$\left|\int_{\mathbb{R}^\mathbf{N}}\partial_z^\alpha\partial_x^\beta\Omega(x,z)\prod_{i=1}^n\varphi(R_iz_i)dz\right| \;\leq\; \mathfrak{B}_{\alpha\,\beta}\left\{1+\sum_{i=1}^n R_i\right\}^{\rho|\beta|+|\alpha|} \tag{1. 6}$$

for every multi-indices α, β and every $R_i>0, i=1,2,\ldots,n$.

Theorem One *Let $\mathbf{T}$ defined in (1. 3) with $\sigma\in\mathbf{S}^0_\rho, 0<\rho<1$ satisfying (1. 2). We have*

$$\left\|\mathbf{T}f\right\|_{\mathbf{L}^p(\mathbb{R}^\mathbf{N})} \;\leq\; \mathfrak{B}_{p\,\rho}\left\|f\right\|_{\mathbf{L}^p(\mathbb{R}^\mathbf{N})}, \qquad 1<p<\infty. \tag{1. 7}$$

Moreover, this class of operators form an algebra under composition.

This paper is organized as follows. In the next section, we introduce a framework of Littlewood-Paley projections and obtain certain combinatorial estimates on the regarding multi-parameter dyadic decomposition. In section 3, we give a characterization between $\sigma\in\mathbf{S}^0_\rho$ and Ω satisfying (1. 5)-(1. 6). In section 4, we prove a fundamental lemma. As a corollary, $\mathbf{T}$ with $\sigma\in\mathbf{S}^0_\rho$ form an algebra. In section 5, we show that every partial operator is bounded by the strong maximal function. Furthermore, these operators enjoy a desired property of almost orthogonality. In section 6, we conclude the $\mathbf{L}^p$-norm inequality in (1. 7). In section 7, we prove the required Littlewood-Paley inequality.

2 Combinatorial estimates on the dyadic decomposition

Let $t_i, i = 1, 2, \ldots, n$ be integers. Denote $\mathbf{t} = (t_1, t_2, \ldots, t_n)$. For $0 < \rho < 1$, we write $q = 1/\rho$ and

$$\mathbf{t}_i \xi \;=\; \left(2^{-qt_i}\xi_1,\ \ldots,\ 2^{-t_i}\xi_i,\ \ldots,\ 2^{-qt_i}\xi_n\right), \quad i = 1, 2, \ldots, n. \tag{2. 1}$$

Consider $\varphi \in C_o^\infty(\mathbb{R})$ such that $\varphi(s) = 1$ if $|s| \leq 1$ and $\varphi(s) = 0$ for $|s| > 2$. We define

$$\delta_{\mathbf{t}}(\xi) \;=\; \prod_{i=1}^{n} \phi\left(\mathbf{t}_i \xi\right), \qquad \phi(\xi) \;=\; \varphi\left(|\xi|\right) - \varphi\left(2|\xi|\right). \tag{2. 2}$$

The support of $\delta_{\mathbf{t}}(\xi)$ is contained in the intersection of n elliptical shells, with different homogeneities of given dilations.

A partial operator $\Delta_{\mathbf{t}}$ is defined by

$$\widehat{\Delta_{\mathbf{t}} f}(\xi) \;=\; \delta_{\mathbf{t}}(\xi)\, \widehat{f}(\xi). \tag{2. 3}$$

Let

$$t_\imath \;=\; \max\left\{t_i : i = 1, 2, \ldots, n\right\}. \tag{2. 4}$$

We partition the set $\{1, 2, \ldots, n\} = \mathbf{I} \cup \mathbf{J}$ for which

$$t_\imath \;\leq\; qt_i - \left(2 + \log_2 n\right), \qquad i \in \mathbf{I} \qquad \text{and} \qquad t_\imath \;>\; qt_i - \left(2 + \log_2 n\right), \qquad i \in \mathbf{J}. \tag{2. 5}$$

Index class H: *We say* $\mathbf{t} \in \mathbf{H}$ *if there exists at least one* $i \in \{1, 2, \ldots n\}$ *such that*

$$t_i \;\geq\; \frac{1}{q-1}\left(2 + \log_2 n\right). \tag{2. 6}$$

Lemma 2.1 *Let* $\xi \in \mathbf{supp}\delta_{\mathbf{t}}(\xi)$*. Suppose* $\mathbf{t} \in \mathbf{H}$*. We have*

$$\begin{aligned} |\xi_i| &\approx 2^{t_i}, \qquad i \in \mathbf{I}, \\ |\xi_i| \;\lesssim\; 2^{t_i}, \qquad |\xi_\imath| &\approx 2^{t_\imath} \approx 2^{qt_i}, \qquad i \in \mathbf{J}. \end{aligned} \tag{2. 7}$$

Proof : Note that $\mathbf{t} \in \mathbf{H}$ satisfying (2. 6) guarantees that the set $\mathbf{I}$ is non-empty. In particular, $\imath \in \mathbf{I}$ because of (2. 4).

Consider $\xi \in \mathbf{supp}\delta_{\mathbf{t}}(\xi)$. We write $\xi = \left(\xi_i, \xi_i^\dagger\right) \in \mathbb{R}^{\mathbf{N}_i} \times \mathbb{R}^{\mathbf{N}-\mathbf{N}_i}$ for $i = 1, 2, \ldots, n$.

By definition of $\delta_{\mathbf{t}}(\xi)$ in (2. 2), we find

$$\begin{aligned} |\xi_i| \;&<\; 2^{t_i+1}, \qquad i = 1, 2, \ldots, n \qquad \text{and} \\ |\xi_i| \;&<\; 2^{qt_j+1}, \qquad \text{for } \; j \neq i. \end{aligned} \tag{2. 8}$$

On the other hand, we either have

$$|\xi_i| \;>\; \frac{2^{t_i-1}}{\sqrt{2}} \qquad \text{or} \qquad \left|\xi_i^\dagger\right| \;>\; \frac{2^{qt_i-1}}{\sqrt{2}}, \qquad i = 1, 2, \ldots, n. \tag{2. 9}$$

Let $i \in \mathbf{I}$ in (2. 5). Suppose $|\xi_i| \leq \frac{2^{t_i-1}}{\sqrt{2}}$. From (2. 9), there is some ξ_j for $j \neq i$ such that

$$|\xi_j| \ > \ \frac{2^{qt_i-1}}{\sqrt{2}} \frac{1}{\sqrt{n-1}}. \tag{2. 10}$$

Together with $|\xi_j| < 2^{t_j+1}$ shown in (2. 8), we must have

$$qt_i - 2 - \frac{1}{2} \log_2 2(n-1) \ < \ t_j. \tag{2. 11}$$

However, $i \in \mathbf{I}$ implies $t_j \leq t_\imath \leq qt_i - \Big(2 + \log_2 n\Big)$. The inequality in (2. 11) cannot be true because

$$(1/2) \log_2 2(n-1) \ < \ \log_2 n \qquad \text{for} \qquad n \geq 2. \tag{2. 12}$$

Therefore, by putting together all estimates above, we find

$$\frac{2^{t_i-1}}{\sqrt{2}} \ < \ |\xi_i| \ < \ 2^{t_i+1}, \qquad i \in \mathbf{I}. \tag{2. 13}$$

Recall $\imath \in \mathbf{I}$. From (2. 13) and the second inequality in (2. 8), we find $2^{t_\imath} \approx |\xi_\imath| < 2^{qt_i+1}$ for $i \neq \imath$. Consider $i \in \mathbf{J}$ in (2. 5) where $qt_i - \Big(2 + \log_2 n\Big) < t_\imath$. We necessarily have

$$2^{t_\imath} \approx |\xi_\imath| \approx 2^{qt_i}, \qquad i \in \mathbf{J}. \tag{2. 14}$$

□

Lemma 2.2 *Let $\sigma \in \mathbf{S}^0_\rho, 0 < \rho < 1$. Suppose $\mathbf{t} \in \mathbf{H}$. We have*

$$\Big|\partial^\alpha_\xi \delta_{\mathbf{t}}(\xi)\sigma(x,\xi)\Big| \ \leq \ \mathfrak{B}_\alpha \prod_{i=1}^n \left(\frac{1}{1 + |\xi_i| + |\xi|^\rho}\right)^{|\alpha_i|} \approx \mathfrak{B}_\alpha \prod_{i=1}^n 2^{-t_i|\alpha_i|} \tag{2. 15}$$

for every multi-index α.

Proof: First, $\delta_{\mathbf{t}}(\xi)$ defined in (2. 2) clearly satisfies

$$\Big|\partial^\alpha_\xi \delta_{\mathbf{t}}(\xi)\Big| \ \leq \ \mathfrak{B}_\alpha \prod_{i=1}^n 2^{-t_i|\alpha_i|} \tag{2. 16}$$

for every multi-index α.

Let $q = 1/\rho$, $t_\imath = \max_{i\in\{1,2,\ldots,n\}} t_i$ and $\mathbf{I} \cup \mathbf{J} = \{1, 2, \ldots, n\}$ defined in (2. 5).

Recall **Lemma 2.1**. We have

$$\begin{aligned} &|\xi_i| \approx 2^{t_i}, \qquad |\xi| \approx |\xi_\imath| \approx 2^{t_\imath} \ \lesssim \ 2^{qt_i}, \qquad i \in \mathbf{I}, \\ &|\xi_i| \ \lesssim \ 2^{t_i}, \qquad |\xi| \approx |\xi_\imath| \approx 2^{t_\imath} \approx 2^{qt_i}, \qquad i \in \mathbf{J} \end{aligned} \tag{2. 17}$$

provided that $\xi \in \mathbf{supp}\delta_{\mathbf{t}}(\xi)\sigma(x,\xi)$ and $\mathbf{t} \in \mathbf{H}$.

From (2. 17), we conclude

$$1 + |\xi_i| + |\xi|^\rho \approx 2^{t_i}, \qquad i = 1, 2, \ldots, n. \tag{2. 18}$$

□

3 Characterization between symbols and kernels

Let $\sigma \in \mathbf{S}^0_\rho, 0 < \rho < 1$ satisfying the differential inequality in (1. 2). Consider

$$\Omega(x,z) = \int_{\mathbb{R}^{\mathbf{N}}} e^{2\pi \mathbf{i} z\cdot\xi}\sigma(x,\xi)d\xi, \qquad z \neq 0. \tag{3. 1}$$

For conventions, we fix some notations as follows.

• Given $\xi_i \in \mathbb{R}^{\mathbf{N}_i}$ and a multi-index α_i, we denote $\xi_i^{\alpha_i} = \prod_{j=1}^{\mathbf{N}_i} \xi_{ij}^{\alpha_{ij}}$ for every $i = 1,2,\ldots,n$. Moreover, $\xi^\alpha = \prod_{i=1}^n \xi_i^{\alpha_i}$.

• Let $\imath \in \{1,2,\ldots,n\}$ such that $|z_\imath| = \max_{i=1,2,\ldots,n} |z_i|$.

• For each $i = 1,2,\ldots,n$, $|z_{i\jmath}| = \max_{j=1,2,\ldots,\mathbf{N}_i} |z_{ij}|$ and $|\xi_{i\jmath}| = \max_{j=1,2,\ldots,\mathbf{N}_i} |\xi_{ij}|$.

• $\gamma = (\gamma_1, \gamma_2, \ldots, \gamma_n)$ is a multi-index where $\gamma_i = \gamma_{i\jmath} \geq 0, i = 1,2,\ldots,n$.

• In particular, we write $\gamma_\imath = \gamma_{\imath\jmath} = \lambda_1 + \lambda_2 + \cdots + \lambda_n$ for which $z_\imath^{\lambda_i} = z_{\imath\jmath}^{\lambda_i}$ and $\partial^{\lambda_i}_{\xi_\imath} = \partial^{\lambda_i}_{\xi_{\imath\jmath}}$ where $\lambda_i \geq 0, i = 1,2,\ldots,n$.

Recall $\delta_{\mathbf{t}}(x)$ defined in (2. 2) and the index class $\mathbf{H}$ defined in (2. 6). Consider

$$\begin{aligned} \partial_z^\alpha \partial_x^\beta \Omega(x,z) &= (2\pi\mathbf{i})^{|\alpha|} \int_{\mathbb{R}^{\mathbf{N}}} e^{2\pi\mathbf{i}z\cdot\xi}\xi^\alpha \partial_x^\beta \sigma(x,\xi)d\xi \\ &= (2\pi\mathbf{i})^{|\alpha|} \sum_{\mathbf{t}\in\mathbf{H}} \partial_z^\alpha\partial_x^\beta \Omega_{\mathbf{t}}(x,z) + (2\pi\mathbf{i})^{|\alpha|}\partial_z^\alpha\partial_x^\beta\Omega^\flat(x,z) \end{aligned} \tag{3. 2}$$

where

$$\partial_z^\alpha\partial_x^\beta\Omega_{\mathbf{t}}(x,z) = \int_{\mathbb{R}^{\mathbf{N}}} e^{2\pi\mathbf{i}z\cdot\xi}\xi^\alpha\delta_{\mathbf{t}}(\xi)\partial_x^\beta\sigma(x,\xi)d\xi \tag{3. 3}$$

and

$$\partial_z^\alpha\partial_x^\beta\Omega^\flat(x,z) = \left(\frac{1}{2\pi\mathbf{i}}\right)^{-|\alpha|}\int_{\mathbb{R}^{\mathbf{N}}} e^{2\pi\mathbf{i}z\cdot\xi}\xi^\alpha\widetilde{\delta}(\xi)\partial_x^\beta\sigma(x,\xi)d\xi, \quad \widetilde{\delta}(\xi) = \sum_{\mathbf{t}\notin\mathbf{H}}\delta_{\mathbf{t}}(\xi). \tag{3. 4}$$

First, we focus on the summation of $\mathbf{t} \in \mathbf{H}$ in (3. 2).

Let γ be the multi-index described as above. Consider

$$\partial_z^\alpha\partial_x^\beta\Omega_{\mathbf{t}}(x,z) = \prod_{i=1}^n z_i^{-\gamma_i}\mathbf{V}_{\mathbf{t}}^{\alpha\,\beta\,\gamma}(z) \tag{3. 5}$$

where

$$\begin{aligned} \mathbf{V}_{\mathbf{t}}^{\alpha\,\beta\,\gamma}(z) &= \prod_{i=1}^n z_i^{\gamma_i}\int_{\mathbb{R}^{\mathbf{N}}} e^{2\pi\mathbf{i}z\cdot\xi}\xi^\alpha\delta_{\mathbf{t}}(\xi)\partial_x^\beta\sigma(x,\xi)d\xi \\ &= z_\imath^{\lambda_\imath}\prod_{i\neq\imath} z_i^{\gamma_i}z_\imath^{\lambda_i}\int_{\mathbb{R}^{\mathbf{N}}} e^{2\pi\mathbf{i}z\cdot\xi}\xi_\imath^{\alpha_\imath}\prod_{i\neq\imath}\xi_i^{\alpha_i}\delta_{\mathbf{t}}(\xi)\partial_x^\beta\sigma(x,\xi)d\xi. \end{aligned} \tag{3. 6}$$

By integration by parts *w.r.t* ξ, we find

$$\mathbf{V}_{\mathbf{t}}^{\alpha\,\beta\,\gamma}(z) = \left(\frac{-1}{2\pi\mathbf{i}}\right)^{|\gamma|}\int_{\mathbb{R}^{\mathbf{N}}} e^{2\pi\mathbf{i}z\cdot\xi}\partial_{\xi_\imath}^{\lambda_\imath}\xi_\imath^{\alpha_\imath}\prod_{i\neq\imath}\partial_{\xi_i}^{\gamma_i}\partial_{\xi_\imath}^{\lambda_i}\xi_i^{\alpha_i}\delta_{\mathbf{t}}(\xi)\partial_x^\beta\sigma(x,\xi)d\xi. \tag{3. 7}$$

Recall that $\sigma(x,\xi)$ satisfies the differential inequality in (1. 2). By using **Lemma 2.2**. we find

$$\left|\mathbf{V}_{\mathbf{t}}^{\alpha\,\beta\,\gamma}(z)\right| \;\leq\; \mathfrak{B}_{\alpha\,\beta\,\gamma}\, 2^{-\lambda_\imath t_\imath+\left(\mathbf{N}_\imath+|\alpha_\imath|+\rho|\beta|\right)t_\imath} \prod_{i\neq\imath} 2^{-\gamma_i t_i-\lambda_\imath t_\imath+(\mathbf{N}_i+|\alpha_i|)t_i}. \tag{3. 8}$$

From **Lemma 2.1**, we have $2^{t_\imath} \lesssim 2^{t_i/\rho}$ for every $i=1,2,\ldots,n$. Therefore, (3. 8) can be further bounded by

$$\mathfrak{B}_{\alpha\,\beta\,\gamma}\, 2^{-\lambda_\imath t_\imath+\left(\mathbf{N}_\imath+|\alpha_\imath|+\rho|\beta|\right)t_\imath} \prod_{i\neq\imath} 2^{-\left(\gamma_i+\rho\lambda_i\right)t_i+(\mathbf{N}_i+|\alpha_i|)t_i}. \tag{3. 9}$$

Choose $\lambda_\imath=0$ if $2^{t_\imath}\leq|z_\imath|^{-1}$ and $\lambda_\imath>\mathbf{N}_\imath+|\alpha_\imath|+\rho|\beta|$ if $2^{t_\imath}>|z_\imath|^{-1}$. We have

$$\begin{aligned}
&|z_\imath|^{-\lambda_\imath}\sum_{t_\imath\geq0} 2^{-\lambda_\imath t_\imath+\left(\mathbf{N}_\imath+|\alpha_\imath|+\rho|\beta|\right)t_\imath}\\
&\leq \sum_{2^{t_\imath}\leq|z_\imath|^{-1}} 2^{\left(\mathbf{N}_\imath+|\alpha_\imath|+\rho|\beta|\right)t_\imath} \;+\; |z_\imath|^{-\lambda_\imath}\sum_{2^{t_\imath}>|z_\imath|^{-1}} 2^{-\lambda_\imath t_\imath+\left(\mathbf{N}_\imath+|\alpha_\imath|+\rho|\beta|\right)t_\imath}\\
&\leq \mathfrak{B}\left(\frac{1}{|z_\imath|}\right)^{\mathbf{N}_\imath+|\alpha_\imath|+\rho|\beta|}.
\end{aligned} \tag{3. 10}$$

Let $\lambda_i=0$ for $i\neq\imath$. Choose $\gamma_i=0$ for $2^{t_i}\leq|z_i|^{-1}$ and $\gamma_i>\mathbf{N}_i+|\alpha_i|$ for $2^{t_i}>|z_i|^{-1}$. We have

$$\begin{aligned}
&|z_i|^{-\gamma_i}\sum_{t_i\geq0} 2^{-\gamma_i t_i+(\mathbf{N}_i+|\alpha_i|)t_i}\\
&\leq \sum_{2^{t_i}\leq|z_i|^{-1}} 2^{(\mathbf{N}_i+|\alpha_i|)t_i} \;+\; |z_i|^{-\gamma_i}\sum_{2^{t_i}>|z_i|^{-1}} 2^{-\gamma_i t_i+(\mathbf{N}_i+|\alpha_i|)t_i}\\
&\leq \mathfrak{B}\left(\frac{1}{|z_i|}\right)^{\mathbf{N}_i+|\alpha_i|}.
\end{aligned} \tag{3. 11}$$

Let $\gamma_i=0$ for every $i\neq\imath$. Choose $\lambda_i=0$ for $2^{t_i}\leq|z_\imath|^{-1/\rho}$ and $\lambda_i>(\mathbf{N}_i+|\alpha_i|)/\rho$ for $2^{t_i}>|z_\imath|^{-1/\rho}$. We have

$$\begin{aligned}
&|z_\imath|^{-\lambda_i}\sum_{t_i\geq0} 2^{-\rho\lambda_i t_i+(\mathbf{N}_i+|\alpha_i|)t_i}\\
&\leq \sum_{2^{t_i}\leq|z_\imath|^{-1/\rho}} 2^{(\mathbf{N}_i+|\alpha_i|)t_i} \;+\; |z_\imath|^{-\lambda_i}\sum_{2^{t_i}>|z_\imath|^{-1/\rho}} 2^{-\rho\lambda_i t_i+(\mathbf{N}_i+|\alpha_i|)t_i}\\
&\leq \mathfrak{B}\left(\frac{1}{|z_\imath|^{1/\rho}}\right)^{\mathbf{N}_i+|\alpha_i|}.
\end{aligned} \tag{3. 12}$$

Recall (3. 5)-(3. 7). By putting together the above estimates from (3. 8) to (3. 12), we find

$$\sum_{\mathbf{t}\in\mathbf{H}}\left|\partial_z^\alpha\partial_x^\beta\Omega_{\mathbf{t}}(x,z)\right| \;\leq\; \mathfrak{B}_{\alpha\,\beta}\left(\frac{1}{|z_\imath|}\right)^{\mathbf{N}_\imath+|\alpha_\imath|+\rho|\beta|}\prod_{i\neq\imath}\left(\frac{1}{|z_i|+|z_\imath|^{1/\rho}}\right)^{\mathbf{N}_i+|\alpha_i|} \tag{3. 13}$$

for $|z|>0$.

Because $|z| \approx |z_\imath| = \max_{i\in\{1,2,\ldots,n\}} |z_i|$, we further have

$$\sum_{\mathbf{t}\in\mathbf{H}} \left|\partial_z^\alpha \partial_x^\beta \Omega_{\mathbf{t}}(x,z)\right| \ \leq\ \mathfrak{B}_{\alpha\ \beta} \left(\frac{1}{|z|}\right)^{\rho|\beta|} \prod_{i=1}^n \left(\frac{1}{|z_i| + |z|^{1/\rho}}\right)^{\mathbf{N}_i+|\alpha_i|}, \qquad |z|>0. \tag{3. 14}$$

On the other hand, by integration by parts *w.r.t* ξ, we find that $\sum_{\mathbf{t}\in\mathbf{H}} \partial_z^\alpha \partial_x^\beta \Omega_{\mathbf{t}}(x,z)$ decays rapidly as $|z| \longrightarrow \infty$.

Return to $\Omega^\flat(x,z)$ defined in (3. 4). Note that $\widetilde{\delta}(\xi) = \sum_{\mathbf{t}\notin\mathbf{H}} \delta_{\mathbf{t}}(\xi)$ has a compact support depending on the index class $\mathbf{H}$ given in (2. 6). By definition of $\delta_{\mathbf{t}}(\xi)$ in (2. 2), we find $|\xi| \lesssim 2^{\frac{\rho}{1-\rho}}$ for $0<\rho<1$ whenever $\xi \in \mathbf{supp}\widetilde{\delta}(\xi)$. This implies

$$\left|\partial_z^\alpha \partial_x^\beta \Omega^\flat(x,z)\right| \ \leq\ \mathfrak{B}_{\alpha\ \beta}\, 2^{\frac{\rho}{1-\rho}\left[\mathbf{N}+|\alpha|+\rho|\beta|\right]}. \tag{3. 15}$$

From (2. 1)-(2. 2), we have

$$\begin{aligned}
\widetilde{\delta}(\xi) &= \sum_{\mathbf{t}\notin\mathbf{H}} \delta_{\mathbf{t}}(\xi) \\
&= \prod_{i=1}^n \varphi\left(\sqrt{2^{-2t/\rho}|\xi_1|^2 + \cdots + 2^{-2t}|\xi_i|^2 + \cdots + 2^{-2t/\rho}|\xi_n|^2}\right)
\end{aligned} \tag{3. 16}$$

where $\varphi \in C_o^\infty(\mathbb{R})$ is the smooth *bump*-function as before and t is the largest integer such that $t < \frac{\rho}{1-\rho}\left(2+\log_2 n\right)$. Every derivative of $\widetilde{\delta}(\xi)$ is either bounded by $\mathfrak{B}2^{-t}$ when $|\xi| \approx 2^t$ or zero otherwise.

Let $\sigma \in \mathbf{S}_\rho$ satisfying (1. 2). Rewrite

$$\partial_z^\alpha \partial_x^\beta \Omega^\flat(x,z) \ =\ \left(\frac{1}{2\pi\mathbf{i}}\right)^{-|\alpha|} \sum_{j\in\mathbb{Z}} \int_{\mathbb{R}^{\mathbf{N}}} e^{2\pi\mathbf{i}z\cdot\xi} \xi^\alpha \widetilde{\delta}(\xi) \partial_x^\beta \sigma(x,\xi) \varphi\left(2^{-j}|\xi|\right) d\xi. \tag{3. 17}$$

An N-fold integration by parts *w.r.t* $\xi_\imath$ shows

$$\begin{aligned}
\left|\int_{\mathbb{R}^{\mathbf{N}}} e^{2\pi\mathbf{i}z\cdot\xi} \xi^\alpha \widetilde{\delta}(\xi) \partial_x^\beta \sigma(x,\xi) \varphi\left(2^{-j}|\xi|\right) d\xi\right| \ &\leq\ \mathfrak{B}_N \left(\frac{1}{|z_\imath|}\right)^N 2^{j\left[\mathbf{N}+|\alpha|+\rho|\beta|\right]} 2^{-j\rho N} \\
&\leq\ \mathfrak{B}_N \left(\frac{1}{|z|}\right)^N 2^{j\left[\mathbf{N}+|\alpha|+\rho|\beta|\right]} 2^{-j\rho N}.
\end{aligned} \tag{3. 18}$$

Choose $N=0$ for $2^j \leq |z|^{-1/\rho}$ and $N > (\mathbf{N}+|\alpha|+\rho|\beta|)/\rho$ for $2^j > |z|^{-1/\rho}$. We have

$$\begin{aligned}
\left|\partial_z^\alpha \partial_x^\beta \Omega^\flat(x,z)\right| \ &\leq\ \mathfrak{B}_{\alpha\ N} \left(\frac{1}{|z|}\right)^N 2^{j\left[\mathbf{N}+|\alpha|+\rho|\beta|\right]} 2^{-j\rho N} \\
&\leq\ \mathfrak{B}_\alpha \sum_{2^j \leq |z|^{-1/\rho}} 2^{j\left[\mathbf{N}+|\alpha|+\rho|\beta|\right]} \ +\ \mathfrak{B}_{\alpha\ N} \left(\frac{1}{|z|}\right)^N \sum_{2^j > |z|^{-1/\rho}} 2^{j\left[\mathbf{N}+|\alpha|+\rho|\beta|\right]} 2^{-j\rho N} \\
&\leq\ \mathfrak{B}_{\alpha\ \beta} \left(\frac{1}{|z|^{1/\rho}}\right)^{\mathbf{N}+|\alpha|+\rho|\beta|}.
\end{aligned} \tag{3. 19}$$

Together with (3. 15), $\partial_z^\alpha\partial_x^\beta\Omega^\flat(x,z)$ satisfies the norm inequality in (3. 19) for $|z|^{-1} \lesssim 2^{\frac{\rho^2}{1-\rho}}$. Compare (3. 19) to (3. 14). From direct computation, we find

$$\left|\partial_z^\alpha\partial_x^\beta\Omega^\flat(x,z)\right| \;\leq\; \mathfrak{B}_{\alpha\,\beta}\, 2^{\rho\left[\mathbf{N}+|\alpha|+\rho|\beta|\right]}\left(\frac{1}{|z|}\right)^{\rho|\beta|}\prod_{i=1}^n\left(\frac{1}{|z_i|+|z|^{1/\rho}}\right)^{\mathbf{N}_i+|\alpha_i|}, \quad |z|>0. \tag{3. 20}$$

From (3. 2) to (3. 20), we conclude (1. 5).

Let $\varphi_i \in C_o^\infty(\mathbb{R}^{\mathbf{N}_i})$, $i=1,2,\ldots,n$ be normalized *bump*-functions. We consider

$$\begin{aligned}\zeta(x,R_1,\ldots,R_n) \;&=\; \int_{\mathbb{R}^\mathbf{N}}\partial_z^\alpha\partial_x^\beta\Omega(x,z)\prod_{i=1}^n\varphi_i\left(R_iz_i\right)dz_i\\ &=\; (-2\pi\mathbf{i})^{|\alpha|}\int_{\mathbb{R}^\mathbf{N}}\partial_x^\beta\sigma(x,\xi)\prod_{i=1}^n R_i^{-\mathbf{N}_i}\widehat{\varphi}_i\left(-R_i^{-1}\xi_i\right)\xi_i^{\alpha_i}d\xi_i.\end{aligned} \tag{3. 21}$$

Define

$$\vartheta_{\alpha\,\beta}(\xi) \;=\; 1+\sum_{i=1}^n|\xi_i|^{\rho|\beta|+|\alpha|}. \tag{3. 22}$$

Rewrite $\zeta(x,R_1,\ldots,R_n)$ in (3. 21) as

$$\begin{aligned}&(-2\pi\mathbf{i})^{|\alpha|}\sum_{i=1}^n\int_{\mathbb{R}^\mathbf{N}}|\xi_i|^{\rho|\beta|+|\alpha|}\left\{\prod_{i=1}^n\xi_i^{\alpha_i}\partial_x^\beta\sigma(x,\xi)\vartheta_{\alpha\,\beta}^{-1}(\xi)\right\}\prod_{i=1}^n R_i^{-\mathbf{N}_i}\widehat{\varphi}_i\left(-R_i^{-1}\xi_i\right)d\xi\\ &+\;(-2\pi\mathbf{i})^{|\alpha|}\int_{\mathbb{R}^\mathbf{N}}\left\{\prod_{i=1}^n\xi_i^{\alpha_i}\partial_x^\beta\sigma(x,\xi)\vartheta_{\alpha\,\beta}^{-1}(\xi)\right\}\prod_{i=1}^n R_i^{-\mathbf{N}_i}\widehat{\varphi}_i\left(-R_i^{-1}\xi_i\right)d\xi.\end{aligned} \tag{3. 23}$$

By changing dilations $\xi_i \longrightarrow R_i\xi_i$ inside (3. 23), we find

$$|\zeta(x,R_1,\ldots,R_n)| \;\leq\; \mathfrak{B}_{\alpha\,\beta}\left\{1+\sum_{i=1}^n R_i\right\}^{\rho|\beta|+|\alpha|}. \tag{3. 24}$$

From (3. 21) to (3. 24), we conclude (1. 6).

Suppose that $\Omega(x,z)$ satisfies (1. 5) and (1. 6). Define

$$\rho_i(\xi) \;=\; 1+|\xi_i|+|\xi|^\rho, \qquad i=1,2,\ldots,n. \tag{3. 25}$$

Consider

$$\partial_\xi^\alpha\partial_x^\beta\sigma(x,\xi)\prod_{i=1}^n\left(\rho_i(\xi)\right)^{|\alpha_i|} \;=\; (-2\pi\mathbf{i})^{|\alpha|}\int_{\mathbb{R}^\mathbf{N}}\partial_x^\beta\Omega(x,z)e^{-2\pi\mathbf{i}z\cdot\xi}\prod_{i=1}^n\left(\rho_i(\xi)\right)^{|\alpha_i|}z_i^{\alpha_i}dz. \tag{3. 26}$$

We aim to show

$$\left|\partial_\xi^\alpha\partial_x^\beta\sigma(x,\xi)\prod_{i=1}^n\left(\rho_i(\xi)\right)^{|\alpha_i|}\right| \;\leq\; \mathfrak{B}_{\alpha\,\beta}\;(1+|\xi|)^{\rho|\beta|} \tag{3. 27}$$

for every multi-indices α,β.

Remark 3.1 *We assume $|\xi|>1$ in the following. The case $|\xi|\le 1$ can be easily deduced from the regarding estimates.*

Let $\phi_i\in C_o^\infty(\mathbb{R}^{\mathbf{N}_i})$, $i=1,2,\ldots,n$ such that

$$\phi_i(z_i) \;=\; 1, \quad |z_i| \;\le\; \frac{1}{2} \qquad \text{and} \qquad \phi_i(z_i) \;=\; 0, \quad |z_i| \;>\; 1. \tag{3. 28}$$

Define

$$\mathbf{P}(x,\xi) \;=\; \int_{\mathbb{R}^{\mathbf{N}}} \partial_x^\beta\Omega(x,z)\prod_{i=1}^n e^{-2\pi\mathbf{i}z_i\cdot\xi_i}\left(\rho_i(\xi)\right)^{|\alpha_i|} z_i^{\alpha_i}\prod_{i=1}^n \phi_i\left(\rho_i(\xi)z_i\right)dz. \tag{3. 29}$$

Observe that

$$z_i^{\alpha_i}\phi_i\left(z_i\right)\mathbf{exp}\left\{-2\pi\mathbf{i}\frac{z_i\cdot\xi_i}{\rho_i(\xi)}\right\}, \qquad i=1,2,\ldots,n \tag{3. 30}$$

is another normalized *bump*-function.

By using (1. 6), we have

$$|\mathbf{P}(x,\xi)| \;\le\; \mathfrak{B}_\beta\left\{1+\sum_{i=1}^n\rho_i(\xi)\right\}^{\rho|\beta|} \;\le\; \mathfrak{B}_\beta\;(1+|\xi|)^{\rho|\beta|}. \tag{3. 31}$$

On the other hand, define

$$\mathbf{E}(x,\xi) \;=\; \int_{\mathbb{R}^{\mathbf{N}}} \partial_x^\beta\Omega(x,z)\prod_{i=1}^n e^{-2\pi\mathbf{i}z_i\cdot\xi_i}\left(\rho_i(\xi)\right)^{|\alpha_i|} z_i^{\alpha_i}\left\{1-\prod_{i=1}^n \phi_i\left(\rho_i(\xi)z_i\right)\right\}dz. \tag{3. 32}$$

Recall the multi-index γ from the beginning of this section. Denote $|\xi_\ell|=\max_{i\in\{1,2,\ldots,n\}}|\xi_i|$. Consider a series of integration by parts as follows.

We integrate by parts *w.r.t* z_ℓ inside (3. 32) and stop whenever there is a partial derivative of $1-\prod_{i=1}^n\varphi_i\left(\rho_i(\xi)z_i\right)$ appeared. The resulting term can be expressed as

$$\begin{aligned}\mathbf{E}_\gamma(x,\xi) \;=\; \mathfrak{C}_\gamma\int_{\mathbb{R}^{\mathbf{N}}}|\xi_\ell|^{-\gamma_\ell}\prod_{i=1}^n\left(\rho_i(\xi)\right)^{|\alpha_i|}\left\{\partial_{z_\ell}^{\gamma_\ell}\partial_x^\beta\Omega(x,z)z_\ell^{\alpha_\ell}\right\}\\ \prod_{i=1}^n e^{-2\pi\mathbf{i}z_i\cdot\xi_i}\prod_{i\ne\ell}z_i^{\alpha_i}|\xi_\ell|^{-1}\partial_{z_\ell}\left\{1-\prod_{i=1}^n\phi_i\left(z_i\right)\right\}dz.\end{aligned} \tag{3. 33}$$

Note that $\prod_{i=1}^n\phi_i\left(\rho_i(\xi)z_i\right)=0$ if $|z_i|>\rho_i^{-1}(\xi)$ for any $i=1,2,\ldots,n$. Therefore, we have $|\xi_\ell|^{-1}\partial_{z_\ell}\left\{1-\prod_{i=1}^n\phi_i\left(\rho_i(\xi)z_i\right)\right\}$ as another product of normalized *bump*-functions having the same characteristic of $\prod_{i=1}^n\phi_i(\rho_i(\xi)z_i)$.

By carrying out a similar argument as (3. 29)-(3. 31) and using (1. 6), we find

$$\left|\mathbf{E}_\gamma(x,\xi)\right| \;\le\; \mathfrak{B}_{\beta\,\gamma}\;|\xi|^{-\gamma_\ell}\left\{1+\sum_{i=1}^n\rho_i(\xi)\right\}^{\rho|\beta|+\gamma_\ell} \;\le\; \mathfrak{B}_{\beta\,\gamma}\;(1+|\xi|)^{\rho|\beta|}. \tag{3. 34}$$

We continue this process until $\gamma_\ell > \sum_{i=1}^n |\alpha_i|$. Our assertion now reduces to

$$
\begin{aligned}
\mathcal{E}(x,\xi) &= \int_{\mathbb{R}^{\mathbf{N}}} |\xi_\ell|^{-\gamma_\ell} \prod_{i=1}^n (\rho_i(\xi))^{|\alpha_i|} \left\{\partial_{z_\ell}^{\gamma_\ell}\partial_x^\beta \Omega(x,z) z_\ell^{\alpha_\ell}\right\} \\
&\quad \prod_{i=1}^n e^{-2\pi \mathbf{i} z_i\cdot\xi_i} \prod_{i\neq\ell} z_i^{\alpha_i} \left\{1-\prod_{i=1}^n \phi_i\left(\rho_i(\xi)z_i\right)\right\} dz \\
&= \int_{\mathbb{R}^{\mathbf{N}}} |\xi_\ell|^{-\gamma_\ell} \prod_{i=1}^n (\rho_i(\xi))^{|\alpha_i|} \left\{\partial_{z_\ell}^{\gamma_\ell}\partial_x^\beta \Omega(x,z) z_\ell^{\alpha_\ell}\right\} \\
&\quad \prod_{i=1}^n e^{-2\pi \mathbf{i} z_i\cdot\xi_i} \prod_{i\neq\ell} z_i^{\alpha_i} \left\{1-\prod_{i=1}^n \phi_i\left(\rho_i(\xi)z_i\right)\right\} \prod_{i=1}^n \phi(z_i) dz \\
&+ \int_{\mathbb{R}^{\mathbf{N}}} |\xi_\ell|^{-\gamma_\ell} \prod_{i=1}^n (\rho_i(\xi))^{|\alpha_i|} \left\{\partial_{z_\ell}^{\gamma_\ell}\partial_x^\beta \Omega(x,z) z_\ell^{\alpha_\ell}\right\} \\
&\quad \prod_{i=1}^n e^{-2\pi \mathbf{i} z_i\cdot\xi_i} \prod_{i\neq\ell} z_i^{\alpha_i} \left\{1-\prod_{i=1}^n \phi_i\left(\rho_i(\xi)z_i\right)\right\} \left\{1-\prod_{i=1}^n \phi(z_i)\right\} dz \\
&\doteq \mathcal{E}_1(x,\xi)+\mathcal{E}_2(x,\xi).
\end{aligned} \tag{3. 35}
$$

Note that $1-\prod_{i=1}^n \phi(z_i)=0$ if $|z_i|\le\frac{1}{2}, i=1,2,\ldots,n$. Hence that

$$
|\mathcal{E}_2(x,\xi)| \;\le\; \mathfrak{B}_{\alpha\,\beta} \tag{3. 36}
$$

provided that $\partial_{z_\ell}^{\gamma_\ell}\partial_x^\beta\Omega(x,z)$ decays rapidly as $|z|\longrightarrow\infty$.

Define

$$
\begin{aligned}
\mathcal{E}_o(x,\xi) &= \int_{\mathbb{R}^{\mathbf{N}}} |\xi_\ell|^{-\gamma_\ell} \prod_{i=1}^n (\rho_i(\xi))^{|\alpha_i|} \left\{\partial_{z_\ell}^{\gamma_\ell}\partial_x^\beta \Omega(x,z) z_\ell^{\alpha_\ell}\right\} \\
&\quad \prod_{i=1}^n e^{-2\pi \mathbf{i} z_i\cdot\xi_i} \prod_{i\neq\ell} z_i^{\alpha_i} \prod_{i=1}^n \phi_i\left(\rho_i(\xi)z_i\right) \prod_{i=1}^n \phi(z_i) dz \\
&= \int_{\mathbb{R}^{\mathbf{N}}} |\xi_\ell|^{-\gamma_\ell} \prod_{i=1}^n (\rho_i(\xi))^{|\alpha_i|} \left\{\partial_{z_\ell}^{\gamma_\ell}\partial_x^\beta \Omega(x,z) z_\ell^{\alpha_\ell}\right\} \\
&\quad \prod_{i=1}^n e^{-2\pi \mathbf{i} z_i\cdot\xi_i} \prod_{i\neq\ell} z_i^{\alpha_i} \prod_{i=1}^n \phi_i\left(\rho_i(\xi)z_i\right) dz.
\end{aligned} \tag{3. 37}
$$

Again, by carrying out a similar argument as (3. 29)-(3. 31) and using (1. 6), we find

$$
|\mathcal{E}_o(x,\xi)| \;\le\; \mathfrak{B}_{\beta\,\gamma}\, |\xi|^{-\gamma_\ell} \left\{1+\sum_{i=1}^n \rho_i(\xi)\right\}^{\rho|\beta|+\gamma_\ell} \;\le\; \mathfrak{B}_{\alpha\,\beta}\, (1+|\xi|)^{\rho|\beta|}. \tag{3. 38}
$$

On the other hand, the norm of

$$\mathcal{E}_3(x,\xi) \;=\; \int_{\mathbb{R}^{\mathbf{N}}} |\xi_\ell|^{-\gamma_\ell} \prod_{i=1}^{n} (\rho_i(\xi))^{|\alpha_i|} \left\{ \partial_{z_\ell}^{\gamma_\ell} \partial_x^{\beta} \Omega(x,z) z_\ell^{\alpha_\ell} \right\} \prod_{i=1}^{n} e^{-2\pi \mathbf{i} z_i\cdot\xi_i} \prod_{i\neq\ell} z_i^{\alpha_i} \prod_{i=1}^{n} \phi_i(z_i)\, dz \tag{3. 39}$$

is bounded by a constant $\mathfrak{B}_{\alpha\ \beta}$. Together with (3. 38), we have

$$|\mathcal{E}_1(x,\xi)| \;=\; |\mathcal{E}_3(x,\xi) - \mathcal{E}_o(x,\xi)| \;\leq\; \mathfrak{B}_{\alpha\ \beta}\ (1+|\xi|)^{\rho|\beta|}. \tag{3. 40}$$

4 A fundamental lemma

Let $0<\rho<1$. Consider $\Theta \in C^\infty(\mathbb{R}^{\mathbf{N}} \times \mathbb{R}^{\mathbf{N}} \times \mathbb{R}^{\mathbf{N}} \times \mathbb{R}^{\mathbf{N}})$ of which

$$\begin{aligned}&\left| \partial_\xi^{\alpha^1} \partial_\eta^{\alpha^2} \partial_x^{\beta^1} \partial_y^{\beta^2} \Theta(x,y,\xi,\eta) \right| \\ &\leq\; \mathfrak{B}_{\alpha^1\ \alpha^2\ \beta^1\ \beta^2}\ \vartheta(\xi,\eta) \prod_{i=1}^{n} \left(\frac{1}{1+|\xi_i|+|\xi|^\rho} \right)^{|\alpha_i^1|} \left(\frac{1}{1+|\eta_i|+|\eta|^\rho} \right)^{|\alpha_i^2|} (1+|\xi|)^{\rho|\beta^1|} (1+|\eta|)^{\rho|\beta^2|}\end{aligned} \tag{4. 1}$$

for every multi-indices $\alpha^1, \alpha^2, \beta^1, \beta^2$.

Define

$$\Lambda(x,\xi) \;=\; \iint_{\mathbb{R}^{\mathbf{N}}\times\mathbb{R}^{\mathbf{N}}} e^{2\pi \mathbf{i}(x-y)\cdot(\xi-\eta)} \Theta(x,y,\xi,\eta) dy d\eta. \tag{4. 2}$$

Lemma One *Let Λ defined in (4. 2). Suppose that $\Theta \in C^\infty(\mathbb{R}^{\mathbf{N}} \times \mathbb{R}^{\mathbf{N}} \times \mathbb{R}^{\mathbf{N}} \times \mathbb{R}^{\mathbf{N}})$ satisfies the differential inequality in (4. 1). We have*

$$\left| \partial_\xi^\alpha \partial_x^\beta \Lambda(x,\xi) \right| \;\leq\; \mathfrak{B}_{\alpha\ \beta\ \rho} \prod_{i=1}^{n} \left(\frac{1}{1+|\xi_i|+|\xi|^\rho} \right)^{|\alpha_i|} (1+|\xi|)^{\rho|\beta|} \tag{4. 3}$$

for every multi-indices α, β.

Corollary 4.1 *Denote σ_i to be the symbol of $\mathbf{T}_i$ for $i=1,2$. Moreover, σ is the symbol of $\mathbf{T}_1 \circ \mathbf{T}_2$. Suppose $\sigma_1, \sigma_2 \in \mathbf{S}^0_\rho$. We have $\sigma \in \mathbf{S}^0_\rho$.*

Proof A direct computation shows

$$\sigma(x,\xi) \;=\; \iint_{\mathbb{R}^{\mathbf{N}}\times\mathbb{R}^{\mathbf{N}}} e^{2\pi \mathbf{i}(x-y)\cdot(\xi-\eta)} \sigma_1(x,\eta)\sigma_2(y,\xi) dy d\eta. \tag{4. 4}$$

Note that $\sigma_1(x,\eta)\sigma_2(y,\xi)$ satisfies the differential inequality in (4. 1). By using **Lemma One**, we have σ in (4. 4) satisfying the differential inequality in (4. 3). □

We develop the proof of **Lemma One** in analogue to the work of Boutet de Monvel [15] and Beals and Fefferman [16].

Let $0 < \rho < 1$. Define a differential operator

$$\mathfrak{D} \ = \ I - \left(\frac{1}{4\pi^2}\right)\left(\frac{1}{1+|\xi|^2} + \frac{1}{1+|\eta|^2}\right)^{\rho} \Delta_y - \left(\frac{1}{4\pi^2}\right)\left(1+|\xi|^2+|\eta|^2\right)^{\rho} \Delta_\eta \tag{4. 5}$$

and the regarding quadratic function

$$\mathbf{Q}(x,y,\xi,\eta) \ = \ 1 + \left(\frac{1}{1+|\xi|^2} + \frac{1}{1+|\eta|^2}\right)^{\rho} |\xi-\eta|^2 + \left(1+|\xi|^2+|\eta|^2\right)^{\rho} |x-y|^2. \tag{4. 6}$$

Denote $\mathbf{L} = \mathbf{Q}^{-1}\mathfrak{D}$. We find

$$\mathbf{L}^N \mathbf{exp}\Big\{2\pi\mathbf{i}(x-y)\cdot(\xi-\eta)\Big\} \ = \ \mathbf{exp}\Big\{2\pi\mathbf{i}(x-y)\cdot(\xi-\eta)\Big\}, \qquad N \geq 1. \tag{4. 7}$$

Remark 4.1 *We momentarily assume that $\Theta(x,y,\xi,\eta)$ has compact support in both y and η.*

By using the identity in (4. 7) and integration by parts *w.r.t* y and η inside (4. 2), we obtain

$$\iint_{\mathbb{R}^{\mathbf{N}}\times\mathbb{R}^{\mathbf{N}}} {}^t\mathbf{L}^N \Theta(x,y,\xi,\eta) e^{2\pi\mathbf{i}(x-y)\cdot(\xi-\eta)} dy d\eta \tag{4. 8}$$

where ${}^t\mathbf{L} = {}^t\mathfrak{D}\mathbf{Q}^{-1}$.

Lemma 4.1 *Suppose that $\Theta \in C^\infty(\mathbb{R}^{\mathbf{N}} \times \mathbb{R}^{\mathbf{N}} \times \mathbb{R}^{\mathbf{N}} \times \mathbb{R}^{\mathbf{N}})$ satisfies (4. 1). We have*

$$\left|{}^t\mathbf{L}^N \Theta(x,y,\xi,\eta)\right| \ \leq \ \mathfrak{B}_N\, \vartheta(\xi,\eta)\left(1 + \frac{|\xi|^2}{1+|\eta|^2} + \frac{|\eta|^2}{1+|\xi|^2}\right)^{\rho N} \mathbf{Q}^{-N}(x,y,\xi,\eta) \tag{4. 9}$$

for every $N \geq 1$.

Proof Denote

$$\mathbf{a}(\xi,\eta) \ = \ \left(\frac{1}{1+|\xi|^2} + \frac{1}{1+|\eta|^2}\right)^{\rho}, \qquad \mathbf{b}(\xi,\eta) \ = \ \left(1+|\xi|^2+|\eta|^2\right)^{\rho}. \tag{4. 10}$$

From (4. 5)-(4. 6), we find

$${}^t\mathbf{L}^N \Theta(x,y,\xi,\eta) \ = \ \left(\frac{1}{\mathbf{Q}} - \Delta_y \frac{\mathbf{a}}{\mathbf{Q}} - \Delta_\eta \frac{\mathbf{b}}{\mathbf{Q}}\right)^N \Theta(x,y,\xi,\eta), \qquad N \geq 1. \tag{4. 11}$$

Observe that

$$\begin{aligned}
\mathbf{a}(\xi,\eta)\Big|\Delta_y \Theta(x,y,\xi,\eta)\Big| \ &\leq \ \mathfrak{B}\, \mathbf{a}(\xi,\eta)\,(1+|\eta|)^{2\rho}\, \vartheta(\xi,\eta) \\
&\leq \ \mathfrak{B} \left(1 + \frac{|\xi|^2}{1+|\eta|^2} + \frac{|\eta|^2}{1+|\xi|^2}\right)^{\rho} \vartheta(\xi,\eta), \\
\mathbf{b}(\xi,\eta)\Big|\Delta_\eta \Theta(x,y,\xi,\eta)\Big| \ &\leq \ \mathfrak{B}\, \mathbf{b}(\xi,\eta)\left(\frac{1}{1+|\eta|}\right)^{2\rho} \vartheta(\xi,\eta) \\
&\leq \ \mathfrak{B} \left(1 + \frac{|\xi|^2}{1+|\eta|^2} + \frac{|\eta|^2}{1+|\xi|^2}\right)^{\rho} \vartheta(\xi,\eta).
\end{aligned} \tag{4. 12}$$

Hence that if all partial derivatives fall on Θ in (4. 11), then (4. 12) implies (4. 9).

Turn to the general case. From (4. 6) and (4. 10), we verify that

$$\left|\partial_\eta^\alpha \mathbf{a}(\xi,\eta)\right| \leq \mathfrak{B}_\alpha \left(\frac{1}{1+|\eta|}\right)^{|\alpha|} \mathbf{a}(\xi,\eta), \qquad \left|\partial_\eta^\alpha \mathbf{b}(\xi,\eta)\right| \leq \mathfrak{B}_\alpha \left(\frac{1}{1+|\xi|+|\eta|}\right)^{|\alpha|} \mathbf{b}(\xi,\eta) \tag{4. 13}$$

for every multi-index α.

Our task can be finished if we show

$$\left|\partial_\eta^\alpha \partial_y^\beta \mathbf{Q}^{-1}(x,y,\xi,\eta)\right| \leq \mathfrak{B}_{\alpha\,\beta}\, \mathbf{Q}^{-1}(x,y,\xi,\eta) \left(\frac{1}{1+|\eta|}\right)^{\rho|\alpha|} (1+|\xi|+|\eta|)^{\rho|\beta|} \tag{4. 14}$$

for every multi-indices α, β.

Note that $\partial_\eta^\alpha \partial_y^\beta \mathbf{Q}^{-1}$ consists a linear combination of

$$\mathbf{Q}^{-1} \prod_k \left[\mathbf{Q}^{-1} \partial_\eta^{\alpha^k} \partial_y^{\beta^k} \mathbf{Q}\right]^k, \qquad |\alpha| = \sum_k |\alpha^k| k, \qquad |\beta| = \sum_k |\beta^k| k. \tag{4. 15}$$

We need the following preliminary estimates:

• Suppose $|\xi-\eta| \leq (1+|\xi|+|\eta|)^\rho$. We necessarily have $|\xi| \approx |\eta|$ and

$$\begin{aligned} \left(\frac{1}{1+|\xi|^2} + \frac{1}{1+|\eta|^2}\right)^\rho |\xi-\eta| &\leq \mathfrak{B} \left(\frac{1}{1+|\eta|}\right)^\rho, \\ \left(\frac{1}{1+|\xi|^2} + \frac{1}{1+|\eta|^2}\right)^\rho &\leq \mathfrak{B} \left(\frac{1}{1+|\eta|}\right)^{2\rho}. \end{aligned} \tag{4. 16}$$

• Suppose $|\xi-\eta| > (1+|\xi|+|\eta|)^\rho$. We have

$$\begin{aligned} &\mathbf{Q}^{-1}(x,y,\xi,\eta) \left(\frac{1}{1+|\xi|^2} + \frac{1}{1+|\eta|^2}\right)^\rho |\xi-\eta| \\ &\leq \mathfrak{B}|\xi-\eta|^{-1} \leq \mathfrak{B} \left(\frac{1}{1+|\eta|}\right)^\rho, \\ &\mathbf{Q}^{-1}(x,y,\xi,\eta) \left(\frac{1}{1+|\xi|^2} + \frac{1}{1+|\eta|^2}\right)^\rho \leq \mathfrak{B}|\xi-\eta|^{-2} \leq \mathfrak{B} \left(\frac{1}{1+|\eta|}\right)^{2\rho}. \end{aligned} \tag{4. 17}$$

• Suppose $|x-y| \leq (1+|\xi|+|\eta|)^{-\rho}$. We have

$$\begin{aligned} \left(1+|\xi|^2+|\eta|^2\right)^\rho |x-y| &\leq \mathfrak{B} (1+|\xi|+|\eta|)^\rho, \\ \left(1+|\xi|^2+|\eta|^2\right)^\rho &\leq \mathfrak{B} (1+|\xi|+|\eta|)^{2\rho}. \end{aligned} \tag{4. 18}$$

• Suppose $|x-y| > (1+|\xi|+|\eta|)^{-\rho}$. We have

$$\begin{aligned} \mathbf{Q}^{-1}(x,y,\xi,\eta) \left(1+|\xi|^2+|\eta|^2\right)^\rho |x-y| &\leq \mathfrak{B}|x-y|^{-1} \leq \mathfrak{B} (1+|\xi|+|\eta|)^\rho, \\ \mathbf{Q}^{-1}(x,y,\xi,\eta) \left(1+|\xi|^2+|\eta|^2\right)^\rho &\leq \mathfrak{B}|x-y|^{-2} \leq \mathfrak{B} (1+|\xi|+|\eta|)^{2\rho}. \end{aligned} \tag{4. 19}$$

From direct computation and by using (4. 16)-(4. 19) together with (4. 13), we find

$$\left|\mathbf{Q}^{-1}\partial_\eta^\alpha\partial_y^\beta\mathbf{Q}(x,y,\xi,\eta)\right| \leq \mathfrak{B}_{\alpha\,\beta}\left(\frac{1}{1+|\eta|}\right)^{\rho|\alpha|}(1+|\xi|+|\eta|)^{\rho|\beta|} \tag{4. 20}$$

for every multi-indices α, β. □

Proof of Lemma One Recall (4. 2). $\partial_\xi^\alpha\partial_x^\beta\Lambda(x,\xi)$ can be expressed as a linear combination of

$$\iint_{\mathbb{R}^{\mathbf{N}}\times\mathbb{R}^{\mathbf{N}}} e^{2\pi\mathbf{i}(x-y)\cdot(\xi-\eta)}\partial_\xi^{\alpha^1}\partial_x^{\beta^1}\partial_\eta^{\alpha^2}\partial_y^{\beta^2}\Theta(x,y,\xi,\eta)dyd\eta \tag{4. 21}$$

where $|\alpha_i| = |\alpha_i^1| + |\alpha_i^2|, i = 1,2,\ldots,n$ and $|\beta| = |\beta^1| + |\beta^2|$.

Recall (4. 7) and ${}^t\mathbf{L} = {}^t\mathfrak{D}\mathbf{Q}^{-1}$ as shown in (4. 8). By integration by parts *w.r.t* y and η, we find (4. 21) equal to

$$\iint_{\mathbb{R}^{\mathbf{N}}\times\mathbb{R}^{\mathbf{N}}} e^{2\pi\mathbf{i}(x-y)\cdot(\xi-\eta)}{}^t\mathbf{L}^N\partial_\xi^{\alpha^1}\partial_x^{\beta^1}\partial_\eta^{\alpha^2}\partial_y^{\beta^2}\Theta(x,y,\xi,\eta)dyd\eta, \qquad N\geq 1. \tag{4. 22}$$

Case One Suppose $|\xi-\eta| > \frac{1}{2}(1+|\xi|+|\eta|)$. We carry out an M-fold integration by parts *w.r.t* y inside (4. 22).

The function $\Delta_y^M\partial_\xi^{\alpha^1}\partial_\eta^{\alpha^2}\partial_x^{\beta^1}\partial_y^{\beta^2}\Theta(x,y,\xi,\eta)/|\xi-\eta|^{2M}$ satisfies the differential inequality in (4. 1) with norm bounded by

$$\begin{aligned}&\mathfrak{B}_{\alpha\,\beta\,M}\,(1+|\xi|+|\eta|)^{2(\rho-1)M}\\ &\prod_{i=1}^n\left(\frac{1}{1+|\xi_i|+|\xi|^\rho}\right)^{|\alpha_i^1|}\left(\frac{1}{1+|\eta_i|+|\eta|^\rho}\right)^{|\alpha_i^2|}(1+|\xi|)^{\rho|\beta^1|}(1+|\eta|)^{\rho|\beta^2|}.\end{aligned} \tag{4. 23}$$

By applying **Lemma 4.1**, we find

$$\begin{aligned}&\left|{}^t\mathbf{L}^N\left(\frac{1}{|\xi-\eta|}\right)^{2M}\Delta_y^M\partial_\xi^{\alpha^1}\partial_\eta^{\alpha^2}\partial_x^{\beta^1}\partial_y^{\beta^2}\Theta(x,y,\xi,\eta)\right|\\ &\leq\mathfrak{B}_{\alpha\,\beta\,M\,N}\left(1+\frac{|\xi|^2}{1+|\eta|^2}+\frac{|\eta|^2}{1+|\xi|^2}\right)^{\rho N}\mathbf{Q}^{-N}(x,y,\xi,\eta)\\ &(1+|\xi|+|\eta|)^{2(\rho-1)M}\prod_{i=1}^n\left(\frac{1}{1+|\xi_i|+|\xi|^\rho}\right)^{|\alpha_i^1|}\left(\frac{1}{1+|\eta_i|+|\eta|^\rho}\right)^{|\alpha_i^2|}(1+|\xi|)^{\rho|\beta^1|}(1+|\eta|)^{\rho|\beta^2|}.\end{aligned} \tag{4. 24}$$

By choosing M sufficiently large, depending on α, β and ρ, we have

$$\begin{aligned}&\iint_{|\xi-\eta|\,>\,\frac{1}{2}(1+|\xi|+|\eta|)} e^{2\pi\mathbf{i}(x-y)\cdot(\xi-\eta)}{}^t\mathbf{L}^N\left(\frac{1}{|\xi-\eta|}\right)^{2M}\Delta_y^M\partial_\xi^{\alpha^1}\partial_x^{\beta^1}\partial_\eta^{\alpha^2}\partial_y^{\beta^2}\Theta(x,y,\xi,\eta)dyd\eta\\ &\leq\mathfrak{B}_{\alpha\,\beta\,\rho\,N}\prod_{i=1}^n\left(\frac{1}{1+|\xi_i|+|\xi|^\rho}\right)^{|\alpha_i|}(1+|\xi|)^{\rho|\beta|}\iint_{\mathbb{R}^{\mathbf{N}}\times\mathbb{R}^{\mathbf{N}}}\mathbf{Q}^{-N}(x,y,\xi,\eta)dyd\eta.\end{aligned} \tag{4. 25}$$

Recall $\mathbf{Q}(x,y,\xi,\eta)$ defined in (4. 6). By changing variables $\eta \longrightarrow \left(1+|\xi|^2\right)^{\rho/2}(\xi-\eta)$ and $y \longrightarrow \left(\frac{1}{1+|\xi|^2}\right)^{\rho/2}(x-y)$, we find

$$\iint_{\mathbb{R}^{\mathbf{N}}\times\mathbb{R}^{\mathbf{N}}} \mathbf{Q}^{-N}(x,y,\xi,\eta)dyd\eta \ \le\ \mathfrak{B} \iint_{\mathbb{R}^{\mathbf{N}}\times\mathbb{R}^{\mathbf{N}}} \left(1+|\eta|^2+|y|^2\right)^{-N} dyd\eta. \tag{4. 26}$$

The integral converges if N is sufficiently large.

Case Two Suppose $|\xi-\eta| \le \frac{1}{2}(1+|\xi|+|\eta|)$. We necessarily have $|\xi|\approx|\eta|$. Let $\mathbf{U}\cup\mathbf{V} = \{1,2,\ldots,n\}$ such that $|\xi_i-\eta_i| \le \frac{1}{2}(1+|\xi_i|+|\eta_i|)$ if $i\in\mathbf{U}$ and $|\xi_i-\eta_i| > \frac{1}{2}(1+|\xi_i|+|\eta_i|)$ if $i\in\mathbf{V}$.

For every $i\in\mathbf{U}$, we have $|\xi_i|\approx|\eta_i|$. Suppose $|\xi_i| \lesssim |\xi|^\rho$ for every $i\in\mathbf{V}$. The function $\partial_\xi^{\alpha^1}\partial_\eta^{\alpha^2}\partial_x^{\beta^1}\partial_y^{\beta^2}\Theta(x,y,\xi,\eta)$ satisfies the differential inequality in (4. 1) with norm bounded by

$$\begin{aligned} &\mathfrak{B}_{\alpha\,\beta} \prod_{i\in\mathbf{U}}\left(\frac{1}{1+|\xi_i|+|\xi|^\rho}\right)^{|\alpha_i|} \prod_{i\in\mathbf{V}}\left(\frac{1}{1+|\xi_i|+|\xi|^\rho}\right)^{|\alpha_i^1|}\left(\frac{1}{1+|\eta_i|+|\eta|^\rho}\right)^{|\alpha_i^2|}(1+|\xi|)^{\rho|\beta|} \\ \le\ &\mathfrak{B}_{\alpha\,\beta} \prod_{i=1}^n\left(\frac{1}{1+|\xi_i|+|\xi|^\rho}\right)^{|\alpha_i|}(1+|\xi|)^{\rho|\beta|}. \end{aligned} \tag{4. 27}$$

On the other hand, assume that there is at least one $i\in\mathbf{V}$ such that $|\xi|^\rho \lesssim |\xi_i|$. Denote $y_{\mathbf{v}}$ to be the projection of y on the subspace $\bigotimes_{i\in\mathbf{V}}\mathbb{R}^{\mathbf{N}_i}$, same for $\xi_{\mathbf{v}}$ and $\eta_{\mathbf{v}}$.

Again, we carry out an M-fold integration by parts *w.r.t* y inside (4. 22). The function $\Delta_{y_{\mathbf{v}}}^M\partial_\xi^{\alpha^1}\partial_\eta^{\alpha^2}\partial_x^{\beta^1}\partial_y^{\beta^2}\Theta(x,y,\xi,\eta)/|\xi_{\mathbf{v}}-\eta_{\mathbf{v}}|^{2M}$ satisfies the differential inequality in (4. 1) with norm bounded by

$$\begin{aligned} &\mathfrak{B}_{\alpha\,\beta\,M}\ (1+|\xi_{\mathbf{v}}|+|\eta_{\mathbf{v}}|)^{-2M}(1+|\xi|^\rho)^{2M} \\ &\prod_{i\in\mathbf{U}}\left(\frac{1}{1+|\xi_i|+|\xi|^\rho}\right)^{|\alpha_i|} \prod_{i\in\mathbf{V}}\left(\frac{1}{1+|\xi_i|+|\xi|^\rho}\right)^{|\alpha_i^1|}\left(\frac{1}{1+|\eta_i|+|\eta|^\rho}\right)^{|\alpha_i^2|}(1+|\xi|)^{\rho|\beta|} \\ \le\ &\mathfrak{B}_{\alpha\,\beta\,M}\ (1+|\xi_{\mathbf{v}}|+|\eta_{\mathbf{v}}|)^{-2M}(1+|\xi|^\rho)^{2M} \\ &\prod_{i\in\mathbf{U}}\left(\frac{1}{1+|\xi_i|+|\xi|^\rho}\right)^{|\alpha_i|} \prod_{i\in\mathbf{V}}\left(\frac{1}{1+|\xi_i|+|\xi|^\rho}\right)^{|\alpha_i^1|}\left(\frac{1}{1+|\xi|^\rho}\right)^{|\alpha_i^2|}(1+|\xi|)^{\rho|\beta|} \\ \le\ &\mathfrak{B}_{\alpha\,\beta} \prod_{i=1}^n\left(\frac{1}{1+|\xi_i|+|\xi|^\rho}\right)^{|\alpha_i|}(1+|\xi|)^{\rho|\beta|} \end{aligned} \tag{4. 28}$$

provided that M is chosen sufficiently large depending on α.

By repeating the estimates in analogue to (4. 24)-(4. 26), we conclude the same result for $|\xi-\eta| \le \frac{1}{2}(1+|\xi|+|\eta|)$.

Our estimates above are independent from the size of **supp**Θ in y and η. The assumption of compactness can be removed by taking the approximation as discussed in **1.3**, chapterVI of Stein [7]. □

5 Point-wise estimates on partial operators

Denote $\mathbf{M}$ to be the strong maximal operator. Recall $\Delta_{\mathbf{t}}$ defined in (2. 3).

Lemma 5.1 *Let $\sigma \in \mathbf{S}^0_\rho, 0 < \rho < 1$. We have*

$$\left|\Delta_{\mathbf{t}}\mathbf{T}f(x)\right| \;\le\; \mathfrak{B}_\rho\, \mathbf{M}f(x) \tag{5. 1}$$

for every $\mathbf{t} \in \mathbf{H}$ satisfying (2. 6).

Proof : A direct computation shows

$$\begin{aligned}\Delta_{\mathbf{t}}\mathbf{T}f(x) &= \int_{\mathbb{R}^{\mathbf{N}}} f(y)\Omega_{\mathbf{t}}(x,y)dy \\ &= \int_{\mathbb{R}^{\mathbf{N}}} f(y)\left\{\int_{\mathbb{R}^{\mathbf{N}}} e^{2\pi\mathbf{i}(x-y)\cdot\xi}\delta_{\mathbf{t}}(\xi)\Lambda(y,\xi)d\xi\right\}dy\end{aligned} \tag{5. 2}$$

where

$$\Lambda(x,\xi) \;=\; \iint_{\mathbb{R}^{\mathbf{N}}\times\mathbb{R}^{\mathbf{N}}} e^{2\pi\mathbf{i}(x-y)\cdot(\xi-\eta)}\sigma(y,\eta)dyd\eta. \tag{5. 3}$$

Let $\sigma \in \mathbf{S}^0_\rho, 0 < \rho < 1$. By applying **Lemma One**, we find that $\Lambda(x,\xi)$ is bounded and satisfies the differential inequality in (4. 3).

Let $z = x - y$ and

$$\mathbf{t}\xi \;=\; \left(2^{-t_1}\xi_1,\, 2^{-t_2}\xi_2, \ldots,\, 2^{-t_n}\xi_n\right), \qquad \mathbf{t}^{-1}\xi \;=\; \left(2^{t_1}\xi_1,\, 2^{t_2}\xi_2, \ldots,\, 2^{t_n}\xi_n\right). \tag{5. 4}$$

By changing dilations $\xi \longrightarrow \mathbf{t}^{-1}\xi$ and $z \longrightarrow \mathbf{t}z$, the integral in (5. 2) can be written as

$$\int_{\mathbb{R}^{\mathbf{N}}} f(x-\mathbf{t}z)\left\{\int_{\mathbb{R}^{\mathbf{N}}} e^{2\pi\mathbf{i}z\cdot\xi}\delta_{\mathbf{t}}\left(\mathbf{t}^{-1}\xi\right)\Lambda(x-\mathbf{t}z,\mathbf{t}^{-1}\xi)d\xi\right\}dz. \tag{5. 5}$$

Let $q = 1/\rho$. By definition of $\delta_{\mathbf{t}}(\xi)$ in (2. 2), the support of

$$\delta_{\mathbf{t}}\left(\mathbf{t}^{-1}\xi\right) \;=\; \prod_{i=1}^n \phi\left(2^{t_1-qt_i}\xi_1, \ldots, \xi_i, \ldots, 2^{t_n-qt_i}\xi_n\right) \tag{5. 6}$$

is contained in a ball with radius equal to 2.

Recall $\mathbf{I}\cup\mathbf{J} = \{1,2,\ldots,n\}$ defined in (2. 5). By integration by parts *w.r.t* ξ, we find

$$\begin{aligned}&\int_{\mathbb{R}^{\mathbf{N}}} e^{2\pi\mathbf{i}z\cdot\xi}\delta_{\mathbf{t}}\left(\mathbf{t}^{-1}\xi\right)\Lambda(x-\mathbf{t}z,\mathbf{t}^{-1}\xi)d\xi \\ &= \left(\frac{1}{2\pi\mathbf{i}z}\right)^{|\alpha|}\int e^{2\pi\mathbf{i}z\cdot\xi}\prod_{i\in\mathbf{I}}\partial^{\alpha_i}_{\xi_i}\prod_{i\in\mathbf{J}}\partial^{\alpha_i}_{\xi_i}\delta_{\mathbf{t}}\left(\mathbf{t}^{-1}\xi\right)\Lambda(x-\mathbf{t}z,\mathbf{t}^{-1}\xi)d\xi\end{aligned} \tag{5. 7}$$

at $z_i \neq 0$, $i = 1,2,\ldots,n$ for every multi-index α.

The integral in (5. 7) has a bounded norm because $\left|\mathbf{supp}\delta_{\mathbf{t}}\left(\mathbf{t}^{-1}\xi\right)\right| \leq \mathfrak{B}2^n$.

Recall **Lemma 2.1**. Given $\mathbf{t} \in \mathbf{H}$ satisfying (2. 6), we have $|\xi_i| \approx 1$ for $i \in \mathbf{I}$ whenever $\xi \in \mathbf{supp}\delta_{\mathbf{t}}\left(\mathbf{t}^{-1}\xi\right)$. In particular, $\imath \in \mathbf{I}$ of which $|\xi_\imath| = \max_{i\in\{1,2,\ldots,n\}} |\xi_i|$ and $2^{t_\imath} \approx 2^{qt_i}$ for $i \in \mathbf{J}$.

From direct computation, we find

$$\begin{aligned}
&\left|\prod_{i\in\mathbf{I}} \partial_{\xi_i}^{\alpha_i} \prod_{i\in\mathbf{J}} \partial_{\xi_i}^{\alpha_i} \delta_{\mathbf{t}}\left(\mathbf{t}^{-1}\xi\right) \Lambda(x-\mathbf{t}z, \mathbf{t}^{-1}\xi)\right| \\
&\leq \mathfrak{B}_\rho \prod_{i\in\mathbf{I}} \left(2^{t_i}\right)^{|\alpha_i|} \left(\frac{1}{1+2^{t_i}|\xi_i| + |\mathbf{t}^{-1}\xi|^\rho}\right)^{|\alpha_i|} \prod_{i\in\mathbf{J}} \left(2^{t_i}\right)^{|\alpha_i|} \left(\frac{1}{1+2^{t_i}|\xi_i| + |\mathbf{t}^{-1}\xi|^\rho}\right)^{|\alpha_i|} \\
&\leq \mathfrak{B}_\rho \prod_{i\in\mathbf{I}} \left(2^{t_i}\right)^{|\alpha_i|} \left(\frac{1}{1+2^{t_i}|\xi_i|}\right)^{|\alpha_i|} \prod_{i\in\mathbf{J}} \left(2^{t_i}\right)^{|\alpha_i|} \left(\frac{1}{1+2^{t_\imath\rho}|\xi_\imath|^\rho}\right)^{|\alpha_i|} \qquad (5.\ 8) \\
&\leq \mathfrak{B}_\rho \prod_{i\in\mathbf{I}} \left(\frac{1}{|\xi_i|}\right)^{|\alpha_i|} \prod_{i\in\mathbf{J}} \left(\frac{1}{|\xi_\imath|}\right)^{\rho|\alpha_i|} \\
&\leq \mathfrak{B}_{\alpha\ \rho}.
\end{aligned}$$

From (5. 7)-(5. 8), we obtain

$$\left|\Delta_{\mathbf{t}}\mathbf{T}f(x)\right| \leq \mathfrak{B}_{\rho\ N} \int |f(x-\mathbf{t}z)| \left(\frac{1}{1+|z|}\right)^N dz, \qquad N \geq 1. \qquad (5.\ 9)$$

Note that $\left(1+|z|\right)^{-N}$ can be approximated by $\sum_{k=0}^\infty a_k \chi_{\mathbf{B}_k}$ where $a_k > 0$ and $\chi_{\mathbf{B}_k}$ is the indicator function on a ball $\mathbf{B}_k$ centered on the origin of $\mathbb{R}^{\mathbf{N}}$ with radius $r_k > 0$. Moreover, $\sum_k a_k|\mathbf{B}_k| < \infty$ provided that N is sufficiently large. From (5. 9), we have

$$\begin{aligned}
\left|\Delta_{\mathbf{t}}\mathbf{T}f(x)\right| &\leq \mathfrak{B}_\rho \int_{\mathbb{R}^{\mathbf{N}}} |f(x-\mathbf{t}z)| \sum_{k=0}^\infty a_k \chi_{\mathbf{B}_k} dz \\
&\leq \mathfrak{B}_\rho \left\{\sum_{k=0}^\infty a_k |\mathbf{B}_k|\right\} \sup_k \frac{1}{|\mathbf{B}_k|} \int_{\mathbf{B}_k} |f(x-\mathbf{t}z)| dz \qquad (5.\ 10) \\
&\leq \mathfrak{B}_\rho\, \mathbf{M}f(x).
\end{aligned}$$

□

Denote $\mathbf{s}$ to be another n-tuple as same as $\mathbf{t}$.

Lemma 5.2 *Suppose $\sigma \in \mathbf{S}_\rho^0$, $0 < \rho < 1$. We have*

$$\left|\Delta_{\mathbf{t}}\mathbf{T}\Delta_{\mathbf{s}}f(x)\right| \leq \mathfrak{B}_\rho \prod_{i=1}^n 2^{-(1-\rho)|t_i - s_i|}\, \mathbf{M}f(x) \qquad (5.\ 11)$$

for every $\mathbf{t}, \mathbf{s} \in \mathbf{H}$ satisfying (2. 6).

Proof Recall (5. 2)-(5. 3). From direct computation, we have

$$\begin{aligned}\Delta_{\mathbf{t}}\mathbf{T}\Delta_{\mathbf{s}}f(x) &= \int_{\mathbb{R}^{\mathbf{N}}} f(y)\Omega_{\mathbf{ts}}(x,y)dy \\ &= \int_{\mathbb{R}^{\mathbf{N}}} f(y)\left\{\int_{\mathbb{R}^{\mathbf{N}}} e^{2\pi\mathbf{i}(x-y)\cdot\xi}\delta_{\mathbf{t}}(\xi)\Lambda_{\mathbf{s}}(y,\xi)d\xi\right\}dy\end{aligned} \tag{5. 12}$$

where

$$\Lambda_{\mathbf{s}}(x,\xi) = \iint_{\mathbb{R}^{\mathbf{N}}\times\mathbb{R}^{\mathbf{N}}} e^{2\pi\mathbf{i}(x-y)\cdot(\xi-\eta)}\delta_{\mathbf{s}}(\eta)\sigma(y,\eta)dyd\eta. \tag{5. 13}$$

In the following paragraph, we write $\mathbf{c} > 0$ for some fixed, suitable constant.

Define $|\mathbf{t}-\mathbf{s}| = \max_{i\in\{1,2,\ldots,n\}}|t_i - s_i|$. Recall **Lemma 2.2**. Note that $\delta_{\mathbf{s}}(\eta)\sigma(y,\eta)$ satisfies the differential inequality in (4. 1). By using **Lemma One**, we find that $\Lambda_{\mathbf{s}}(x,\xi)$ is bounded and satisfies the differential inequality in (4. 3). For $|\mathbf{t}-\mathbf{s}| \leq \mathbf{c}$, the situation can be handled as **Lemma 5.1**.

From now on, we consider $|\mathbf{t}-\mathbf{s}| > \mathbf{c}$. There is at least one $i \in \{1,2,\ldots,n\}$ such that $|t_i - s_i| > \mathbf{c}$. By integration by parts *w.r.t* y_ℓ, we find

$$\Lambda_{\mathbf{s}}(x,\xi) = \frac{-1}{4\pi^2}\iint_{\mathbb{R}^{\mathbf{N}}\times\mathbb{R}^{\mathbf{N}}} e^{2\pi\mathbf{i}(x-y)\cdot(\xi-\eta)}\delta_{\mathbf{s}}(\eta)\Delta_{y_i}\sigma(y,\eta)\frac{1}{|\xi_i-\eta_i|^2}dyd\eta. \tag{5. 14}$$

Note that the boundary terms are vanished. This can be seen by integration by parts *w.r.t* η whereas $\delta_{\mathbf{s}}(\eta)\Delta_{y_i}\sigma(y,\eta)/|\xi_i-\eta_i|^2$ has a compact support in η for every $\mathbf{s}$.

Let $\sigma \in \mathbf{S}^0_\rho, 0 < \rho < 1$ satisfying (1. 2) and $q = 1/\rho$. Given $\mathbf{t},\mathbf{s} \in \mathbf{H}$ as (2. 6), we consider $\xi \in \mathbf{supp}\delta_{\mathbf{t}}(\xi)$ and $\eta \in \mathbf{supp}\delta_{\mathbf{s}}(\eta)$.

Denote $t_\imath = \max_{i\in\{1,2,\ldots,n\}} t_i$ and $s_\jmath = \max_{i\in\{1,2,\ldots,n\}} s_i$. We define $\mathbf{I}_1\cup\mathbf{J}_1 = \mathbf{I}_2\cup\mathbf{J}_2 = \{1,2,\ldots,n\}$ respectively for $\mathbf{t}$ and $\mathbf{s}$ as (2. 5).

Recall **Lemma 2.1**. We have $|\xi_i|\approx 2^{t_i}, i\in\mathbf{I}_1$ and $|\xi_i| \lesssim 2^{t_i}, 2^{t_\imath}\approx 2^{qt_i}, i\in\mathbf{J}_1$. On the other hand, we have $|\eta_i|\approx 2^{s_i}, i\in\mathbf{I}_2$ and $|\eta_i|\lesssim 2^{s_i}, 2^{s_\jmath}\approx 2^{qs_i}, i\in\mathbf{J}_2$.

Remark 5.1 *We must have $2^{t_i} \lesssim 2^{qt_j}$ and $2^{s_i} \lesssim 2^{qs_j}$ for every $i,j \in \{1,2,\ldots,n\}$ provided that* $\mathbf{supp}\delta_{\mathbf{t}}(\xi) \neq \emptyset$ *and* $\mathbf{supp}\delta_{\mathbf{s}}(\eta) \neq \emptyset$.

1. Suppose $i \in \mathbf{I}_1\cap\mathbf{I}_2$. We have $|\xi_i|\approx 2^{t_i}$, $|\eta_i|\approx 2^{s_i}$ and therefore

$$|\xi_i - \eta_i| \approx \begin{cases} 2^{t_i} = 2^{s_i}2^{(t_i-s_i)}, & t_i > s_i + \mathbf{c}, \\ 2^{s_i} = 2^{t_i}2^{(s_i-t_i)}, & t_i < s_i - \mathbf{c}. \end{cases} \tag{5. 15}$$

From **Remark 5.1**, the first equality in (5. 15) further implies

$$\frac{1+|\eta|}{|\xi_i-\eta_i|^q} \lesssim 2^{-q|t_i-s_i|}. \tag{5. 16}$$

For $t_i < s_i - \mathbf{c}$ and $qt_i \geq s_\jmath - \mathbf{c}$, the second equality in (5. 15) implies (5. 16) again.

For $qt_i < s_\jmath - \mathbf{c}$, **Remark 5.1** implies $|\xi_\jmath| \leq \frac{1}{2}|\eta_\jmath|$. Replace i with $\jmath$ inside (5. 14). We find

$$\frac{1+|\eta|}{|\xi_\jmath - \eta_\jmath|^q} \lesssim 2^{-(q-1)s_\jmath} \lesssim 2^{-(q-1)s_i}. \tag{5. 17}$$

2. Suppose $i \in \mathbf{I}_1 \cap \mathbf{J}_2$. We have $|\xi_i| \approx 2^{t_i}$ and $|\eta_i| \lesssim 2^{s_i}$ where $|\eta| \approx |\eta_\jmath| \approx 2^{s_\jmath} \approx 2^{qs_i}$.

For $t_i > s_\ell - \mathbf{c}$, the first equality in (5. 15) remains valid which further implies (5. 16).

For $t_i < s_\ell - \mathbf{c}$, **Remark 5.1** implies $|\xi_\jmath| \leq \frac{1}{2}|\eta_\jmath|$. Replace i with $\jmath$ inside (5. 14). We find (5. 17).

3. Suppose $i \in \mathbf{J}_1 \cap \mathbf{I}_2$. We have $|\xi_i| \lesssim 2^{t_i}$ where $|\xi| \approx |\xi_\imath| \approx 2^{t_\imath} \approx 2^{qt_i}$ and $|\eta_i| \approx 2^{s_i}$.

For $t_i > s_i + \mathbf{c}$, **Remark 5.1** implies $|\eta_\imath| \leq \frac{1}{2}|\xi_\imath|$. Replace i with $\imath$ inside (5. 14). We find

$$\frac{1+|\eta|}{|\xi_\imath - \eta_\imath|^q} \lesssim 2^{-(q-1)t_\imath} \lesssim 2^{-(q-1)t_i}. \tag{5. 18}$$

Consider $t_i < s_i - \mathbf{c}$ and $qt_i \geq s_\jmath - \mathbf{c}$. Note that the second equality in (5. 15) remains valid which implies (5. 16).

For $qt_i < s_\jmath - \mathbf{c}$, **Remark 5.1** implies $|\xi_\jmath| \leq \frac{1}{2}|\eta_\jmath|$. Replace i with $\jmath$ inside (5. 14). We find (5. 17).

4. Suppose $i \in \mathbf{J}_1 \cap \mathbf{J}_2$. We have $|\xi| \approx 2^{qt_i} \approx 2^{t_\imath}$ and $|\eta| \approx 2^{qs_i} \approx 2^{s_\jmath}$.

For $t_i > s_i + \mathbf{c}$, we have $|\eta_\imath| \leq \frac{1}{2}|\xi_\imath|$. Replace i with $\imath$ inside (5. 14). We find (5. 18).

For $t_i < s_i - \mathbf{c}$, we have $|\xi_\jmath| \leq \frac{1}{2}|\eta_\jmath|$. Replace i with $\jmath$ inside (5. 14). We find (5. 17).

In summary of step **1-4**, we integrate by parts *w.r.t* i, $\imath$ and $\jmath$ depending on different cases. Moreover, the following estimates hold accordingly:

$$\begin{gathered}
|\xi_i - \eta_i| \approx 2^{t_i}, \quad |\xi_i| \approx 2^{t_i}, \quad i \in \mathbf{I}_1, \qquad \text{or} \qquad |\xi_i - \eta_i| \approx 2^{s_i}, \quad |\eta_i| \approx 2^{s_i}, \quad i \in \mathbf{I}_2. \\
|\xi_\imath - \eta_\imath| \approx 2^{t_\imath}, \quad |\xi_\imath| \approx 2^{t_\imath} \qquad \text{and} \qquad |\xi_\jmath - \eta_\jmath| \approx 2^{s_\jmath}, \quad |\eta_\jmath| \approx 2^{s_\jmath}.
\end{gathered} \tag{5. 19}$$

Note that $|\xi_i| \gtrsim |\xi|^\rho$ because $|\xi| \approx |\xi_\imath| \approx 2^{t_\imath} \lesssim 2^{qt_i}$ for $i \in \mathbf{I}_1$. Similarly, $|\eta_i| \gtrsim |\eta|^\rho$ because $|\eta| \approx |\eta_\jmath| \approx 2^{t_\jmath} \lesssim 2^{qt_i}$ for $i \in \mathbf{I}_2$.

We carry out the integration by parts on every i such that $|t_i - s_i| > \mathbf{c}$. The resulting function having an expression of $\delta_{\mathbf{s}}(\eta) \prod \Delta_{y_i}\sigma(y,\eta)/|\xi_i - \eta_i|^2 \prod \Delta_{y_\imath}\sigma(y,\eta)/|\xi_\imath - \eta_\imath|^2 \prod \Delta_{y_\jmath}\sigma(y,\eta)/|\xi_\jmath - \eta_\jmath|^2$ satisfies the differential inequality in (4. 1).

In particular, its norm is bounded by

$$\mathfrak{B} \prod_{i=1}^n 2^{-\left(\frac{q-1}{q}\right)|t_i - s_i|} = \mathfrak{B} \prod_{i=1}^n 2^{-(1-\rho)|t_i - s_i|}. \tag{5. 20}$$

We finish the proof by repeating the estimates from (5. 5) to (5. 10), with $\Omega_{\mathbf{t}}$ replaced by $\Omega_{\mathbf{ts}}$ given in (5. 12). □

6 Conclusion on the $\mathbf{L}^p$-boundedness

Let $\Delta_{\mathbf{t}}$ defined in (2. 2)-(2. 3). For $f \in \mathbf{L}^p(\mathbb{R}^{\mathbf{N}})$ and $g \in \mathbf{L}^{\frac{p}{p-1}}(\mathbb{R}^{\mathbf{N}})$, we consider

$$\int_{\mathbb{R}^{\mathbf{N}}} \mathbf{T}f(x)g(x)dx \;=\; \int_{\mathbb{R}^{\mathbf{N}}} \left\{\sum_{\mathbf{t}} \Delta_{\mathbf{t}}\mathbf{T}f(x)\right\}\left\{\sum_{\mathbf{s}} \Delta_{\mathbf{s}}g(x)\right\}dx. \tag{6. 1}$$

Note that $\mathbf{supp}\delta_{\mathbf{t}}(\xi)\overline{\delta_{\mathbf{s}}(\xi)} \neq \emptyset$ only if $|\mathbf{t}-\mathbf{s}| \leq 2$. Because of Plancherel theorem, our assertion on (6. 1) can be reduced to

$$\int_{\mathbb{R}^{\mathbf{N}}} \sum_{\mathbf{t}} \Delta_{\mathbf{t}}\mathbf{T}f(x)\Delta_{\mathbf{t}}g(x)dx. \tag{6. 2}$$

Let $h_i \in \mathbb{Z}, i = 1, 2, \ldots, n$. We define the n-tuple $(\mathbf{t}+\mathbf{h})_i$ by simply replacing t_i with $t_i + h_i$ respectively in (2. 1). For each $\mathbf{t}$ fixed, we write $f = \sum_{\mathbf{h}} \Delta_{\mathbf{t}+\mathbf{h}}f$. The expression in (6. 2) equals

$$\sum_{\mathbf{h}} \int_{\mathbb{R}^{\mathbf{N}}} \sum_{\mathbf{t}} \Delta_{\mathbf{t}}\mathbf{T}\Delta_{\mathbf{t}+\mathbf{h}}f(x)\Delta_{\mathbf{t}}g(x)dx. \tag{6. 3}$$

Remark 6.1 *We assume* $\mathbf{t}, \mathbf{t}+\mathbf{h} \in \mathbf{H}$ *satisfying (2. 6) in the summation of (6. 3).*

Note that $\widetilde{\delta}(\xi) = \sum_{\mathbf{t}\notin\mathbf{H}} \delta_{\mathbf{t}}(\xi)$ as shown in (3. 16) has a compact support for $|\xi| \lesssim 2^{\frac{\rho}{1-\rho}}$ for $\xi \in \mathbf{supp}\widetilde{\delta}(\xi)$. The symbols of $\mathbf{T}\left(I - \sum_{\mathbf{t}+\mathbf{h}\notin\mathbf{H}} \Delta_{\mathbf{t}+\mathbf{h}}\right)$ and $\left[\left(I - \sum_{\mathbf{t}\notin\mathbf{H}} \Delta_{\mathbf{t}}\right)\mathbf{T}\right]^*$ respectively are $\widetilde{\delta}(\xi)\sigma(x,\xi)$ and $\overline{\widetilde{\delta}(\xi)\Lambda(x,\xi)}$ where $\Lambda(x,\xi)$ is given in (5. 3). Both $\sigma \in \mathbf{S}^0_\rho, 0 < \rho < 1$ and $\Lambda(x,\xi)$ satisfy the differential inequality in (1. 2).

From direct computation, we find

$$\left|\partial_\xi^\alpha \partial_x^\beta \widetilde{\delta}(\xi)\sigma(x,\xi)\right| \;\leq\; \mathfrak{B}_{\alpha\;\beta}\, 2^{\rho|\alpha|}\left(\frac{1}{1+|\xi|}\right)^{|\alpha|}(1+|\xi|)^{\rho|\beta|} \tag{6. 4}$$

for every multi-indices α, β. The same is true for $\overline{\widetilde{\delta}(\xi)\Lambda(x,\xi)}$.

The regarding pseudo differential operators, as well as their adjoint operators, are well known to be bounded on $\mathbf{L}^p$-spaces for $1 < p < \infty$.

By using Schwarz inequality and then Hölder inequality, we have

$$\begin{aligned}
\int_{\mathbb{R}^{\mathbf{N}}} \mathbf{T}f(x)g(x)dx \;&\leq\; \mathfrak{B}\sum_{\mathbf{h}} \int \left\{\sum_{\mathbf{t}} \left(\Delta_{\mathbf{t}}\mathbf{T}\Delta_{\mathbf{t}+\mathbf{h}}f\right)^2(x)\right\}^{\frac{1}{2}} \left\{\sum_{\mathbf{t}} \left(\Delta_{\mathbf{t}}g\right)^2(x)\right\}^{\frac{1}{2}} dx \\
&\leq\; \mathfrak{B}\sum_{\mathbf{h}} \left\|\left\{\sum_{\mathbf{t}} \left(\Delta_{\mathbf{t}}\mathbf{T}\Delta_{\mathbf{t}+\mathbf{h}}f\right)^2\right\}^{\frac{1}{2}}\right\|_{\mathbf{L}^p(\mathbb{R}^{\mathbf{N}})} \left\|\left\{\sum_{\mathbf{t}} \left(\Delta_{\mathbf{t}}g\right)^2\right\}^{\frac{1}{2}}\right\|_{\mathbf{L}^{\frac{p}{p-1}}(\mathbb{R}^{\mathbf{N}})}.
\end{aligned} \tag{6. 5}$$

Lemma 6.1 *Let* $\Delta_{\mathbf{t}}$ *defined in (2. 3). We have*

$$\left\|\left\{\sum_{\mathbf{t}} \left(\Delta_{\mathbf{t}}f\right)^2\right\}^{\frac{1}{2}}\right\|_{\mathbf{L}^p(\mathbb{R}^{\mathbf{N}})} \;\leq\; \mathfrak{B}_p \left\|f\right\|_{\mathbf{L}^p(\mathbb{R}^{\mathbf{N}})}, \qquad 1 < p < \infty. \tag{6. 6}$$

By taking the supremum of all $\|g\|_{\mathbf{L}^{\frac{p}{p-1}}(\mathbb{R}^{\mathbf{N}})} = 1$ in (6. 5) and applying **Lemma 6.1**, we find

$$\left\|\mathbf{T}f\right\|_{\mathbf{L}^p(\mathbb{R}^{\mathbf{N}})} \ \leq\ \mathfrak{B}_p \sum_{\mathbf{h}} \left\| \left\{ \sum_{\mathbf{t}} \left(\Delta_{\mathbf{t}}\mathbf{T}\Delta_{\mathbf{t+h}}f\right)^2 \right\}^{\frac{1}{2}} \right\|_{\mathbf{L}^p(\mathbb{R}^{\mathbf{N}})}, \qquad 1<p<\infty. \tag{6. 7}$$

Recall **Lemma 5.1**. The norm of $\Delta_{\mathbf{t}}\mathbf{T}f$ is bounded by $\mathbf{M}f$ for every $\mathbf{t}\in\mathbf{H}$. By applying the vector-valued inequality of strong maximal function, established by Fefferman and Stein [5] and then using the Littlewood-Paley inequality in (6. 6), we have

$$\begin{aligned} \left\| \left\{ \sum_{\mathbf{t}} \left(\Delta_{\mathbf{t}}\mathbf{T}\Delta_{\mathbf{t+h}}f\right)^2 \right\}^{\frac{1}{2}} \right\|_{\mathbf{L}^p(\mathbb{R}^{\mathbf{N}})} \ &\leq\ \mathfrak{B}\ \left\| \left\{ \sum_{\mathbf{t}} \left(\mathbf{M}\Delta_{\mathbf{t+h}}f\right)^2 \right\}^{\frac{1}{2}} \right\|_{\mathbf{L}^p(\mathbb{R}^{\mathbf{N}})} \\ &\leq\ \mathfrak{B}_p\ \left\| \left\{ \sum_{\mathbf{t}} \left(\Delta_{\mathbf{t+h}}f\right)^2 \right\}^{\frac{1}{2}} \right\|_{\mathbf{L}^p(\mathbb{R}^{\mathbf{N}})} \ \leq\ \mathfrak{B}_p\ \left\|f\right\|_{\mathbf{L}^p(\mathbb{R}^{\mathbf{N}})} \end{aligned} \tag{6. 8}$$

for every $1<p<\infty$.

On the other hand, we have $(\Delta_{\mathbf{t}}\mathbf{T}\Delta_{\mathbf{s}})^* f = \Delta_{\mathbf{s}}^*\mathbf{T}^*\Delta_{\mathbf{t}}^* f = 0$. provided that $\mathbf{supp}\,\delta_{\mathbf{t}}(\xi)\overline{\delta_{\mathbf{s}}(\xi)} = \emptyset$ if $|\mathbf{t}-\mathbf{s}|>2$. Given $\mathbf{h}$ fixed, by applying Cotlar-Stein lemma together with **Lemma 6.1**, we have

$$\left\| \left\{ \sum_{\mathbf{t}} \left(\Delta_{\mathbf{t}}\mathbf{T}\Delta_{\mathbf{t+h}}f\right)^2 \right\}^{\frac{1}{2}} \right\|_{\mathbf{L}^2(\mathbb{R}^{\mathbf{N}})} \ \leq\ \mathfrak{B}_\rho\ \prod_{i=1}^n 2^{-(1-\rho)|h_i|} \left\|f\right\|_{\mathbf{L}^2(\mathbb{R}^{\mathbf{N}})}. \tag{6. 9}$$

From (6. 8) and (6. 9), Riesz-Thorin interpolation theorem implies

$$\left\| \left\{ \sum_{\mathbf{t}} \left(\Delta_{\mathbf{t}}\mathbf{T}\Delta_{\mathbf{t+h}}f\right)^2 \right\}^{\frac{1}{2}} \right\|_{\mathbf{L}^p(\mathbb{R}^{\mathbf{N}})} \ \leq\ \mathfrak{B}_{\rho\ p}\ \prod_{i=1}^n 2^{-\varepsilon|h_i|} \left\|f\right\|_{\mathbf{L}^p(\mathbb{R}^{\mathbf{N}})} \tag{6. 10}$$

for $p\neq 2, 1<p<\infty$ where $\varepsilon=\varepsilon(\rho,p)>0$. By summing over all the $h_i, i=1,2,\ldots,n$ in (6. 7), we obtain the desired result.

7 Proof of the Littlewood-Paley inequality

In order to prove **Lemma 6.1**, we momentarily consider $n=2$. Let $(x,y)\in\mathbb{R}^{\mathbf{N}_1}\times\mathbb{R}^{\mathbf{N}_2}$ and $(\xi,\eta)\in\mathbb{R}^{\mathbf{N}_1}\times\mathbb{R}^{\mathbf{N}_2}$.

Recall ϕ defined in (2. 2). For $0<\rho<1$ and $\delta>0$, we write

$$\widehat{\Phi}_\delta(\xi,\eta) \ =\ \phi\left(\delta\xi, \delta^{\frac{1}{\rho}}\eta\right). \tag{7. 1}$$

Note that

$$\iint_{\mathbb{R}^{\mathbf{N}_1}\times\mathbb{R}^{\mathbf{N}_2}} \Phi_\delta(x,y)dxdy \ =\ \phi(0,0) \ =\ 0, \qquad \delta>0. \tag{7. 2}$$

Moreover, $\Phi(x,y) \doteq \Phi_1(x,y)$ is smooth, bounded and decays rapidly as $|(x,y)| \longrightarrow \infty$.

The regarding square function is defined by

$$\mathfrak{S}_\Phi f(x,y) \;=\; \left\{\int_0^\infty \left(f * \Phi_\delta\right)^2(x,y)\frac{d\delta}{\delta}\right\}^{\frac{1}{2}}. \tag{7. 3}$$

We aim to show

$$\left\|\mathfrak{S}_\Phi f\right\|_{\mathbf{L}^p(\mathbb{R}^\mathbf{N})} \;\leq\; \mathfrak{B}_p \;\left\|f\right\|_{\mathbf{L}^p(\mathbb{R}^\mathbf{N})}, \qquad 1<p<\infty. \tag{7. 4}$$

The kernel of $\mathfrak{S}_\Phi$, denoted by

$$\Omega(x,y) \;=\; \Phi_\delta(x,y) \;=\; \left(\frac{1}{\delta}\right)^{\mathbf{N}_1+\mathbf{N}_2/\rho}\Phi\left(\frac{x}{\delta},\frac{y}{\delta^{1/\rho}}\right) \tag{7. 5}$$

has a Hilbert space valued norm

$$\left|\Omega(x,y)\right|_H \;=\; \left\{\int_0^\infty \left|\Phi_\delta(x,y)\right|^2\frac{d\delta}{\delta}\right\}^{\frac{1}{2}}. \tag{7. 6}$$

Lemma 7.1

$$\left|\partial_x^\alpha\partial_y^\beta\Omega(x,y)\right|_H \;\leq\; \mathfrak{B}_{\alpha\,\beta}\left(\frac{1}{|x|+|y|^\rho}\right)^{\mathbf{N}_1+|\alpha|}\left(\frac{1}{|y|+|x|^{1/\rho}}\right)^{\mathbf{N}_2+|\beta|} \tag{7. 7}$$

for every multi-indices α, β.

Proof 1. Suppose $|x|\leq\delta$ and $|y|\leq\delta^{1/\rho}$. Because Φ is smooth and bounded, (7. 5) implies

$$\left|\Omega(x,y)\right|_H \;\leq\; \mathfrak{B}\left\{\int_{|x|}^\infty\left(\frac{1}{|x|}\right)^{2\mathbf{N}_1+2\mathbf{N}_2/\rho-1}\frac{d\delta}{\delta^2}\right\}^{\frac{1}{2}} \;\leq\; \mathfrak{B}\left(\frac{1}{|x|}\right)^{\mathbf{N}_1+\mathbf{N}_2/\rho} \tag{7. 8}$$

and

$$\left|\Omega(x,y)\right|_H \;\leq\; \mathfrak{B}\left\{\int_{|y|^\rho}^\infty\left(\frac{1}{|y|}\right)^{2\rho\mathbf{N}_1+2\mathbf{N}_2-1}\frac{d\delta}{\delta^{1+1/\rho}}\right\}^{\frac{1}{2}} \;\leq\; \mathfrak{B}\left(\frac{1}{|y|}\right)^{\rho\mathbf{N}_1+\mathbf{N}_2}. \tag{7. 9}$$

2. Suppose $|x|>\delta$ and $|y|\leq\delta^{1/\rho}$. We necessarily have $|x|>|y|^\rho$. Because $\Phi(x,y)$ decays rapidly as $|(x,y)|\longrightarrow\infty$, we have

$$\left|\Omega(x,y)\right|_H \;\leq\; \mathfrak{B}\left\{\int_0^{|x|}\left(\frac{1}{|x|}\right)^{2\mathbf{N}_1+2\mathbf{N}_2/\rho+2}\delta d\delta\right\}^{\frac{1}{2}} \;\leq\; \mathfrak{B}\left(\frac{1}{|x|}\right)^{\mathbf{N}_1+\mathbf{N}_2/\rho}. \tag{7. 10}$$

On the other hand, the estimate in (7. 9) shows

$$\left|\Omega(x,y)\right|_H \;\leq\; \mathfrak{B}\left(\frac{1}{|y|}\right)^{\rho\mathbf{N}_1+\mathbf{N}_2}. \tag{7. 11}$$

3. Suppose $|x|\leq\delta$ and $|y|>\delta^{1/\rho}$. We necessarily have $|x|<|y|^\rho$. The estimate in (7. 8) shows

$$\left|\Omega(x,y)\right|_H \;\leq\; \mathfrak{B}\left(\frac{1}{|x|}\right)^{\mathbf{N}_1+\mathbf{N}_2/\rho}. \tag{7. 12}$$

On the other hand, because $\Phi(x,y)$ decays rapidly as $|(x,y)|\longrightarrow\infty$, we have

$$\Big|\Omega(x,y)\Big|_H ~\leq~ \mathfrak{B}\left\{\int_0^{|y|^\rho}\left(\frac{1}{|y|}\right)^{2\rho\mathbf{N}_1+2\mathbf{N}_2+2}\delta^{2/\rho-1}d\delta\right\}^{\frac{1}{2}} ~\leq~ \mathfrak{B}\left(\frac{1}{|y|}\right)^{\rho\mathbf{N}_1+\mathbf{N}_2}. \tag{7. 13}$$

4. Suppose $|x|>\delta$ and $|y|>\delta^{1/\rho}$. From (7. 10) and (7. 13), we find

$$\Big|\Omega(x,y)\Big|_H ~\leq~ \mathfrak{B}\left(\frac{1}{|x|}\right)^{\mathbf{N}_1+\mathbf{N}_2/\rho}, \qquad \Big|\Omega(x,y)\Big|_H ~\leq~ \mathfrak{B}\left(\frac{1}{|y|}\right)^{\rho\mathbf{N}_1+\mathbf{N}_2}. \tag{7. 14}$$

All together, we obtain

$$\Big|\Omega(x,y)\Big|_H ~\leq~ \mathfrak{B}\left(\frac{1}{|x|+|y|^\rho}\right)^{\mathbf{N}_1}\left(\frac{1}{|y|+|x|^{1/\rho}}\right)^{\mathbf{N}_2}. \tag{7. 15}$$

Observe that every ∂_x acting on Φ_δ gains a constant multiple of δ^{-1} and every ∂_y acting on Φ_δ gains a constant multiple of $\delta^{-1/\rho}$ respectively. By carrying out the same estimates as above, we prove the differential inequality in (7. 7). □

Define

$$\lambda(x,y) ~=~ \max\Big\{|x|,|y|^\rho\Big\} \tag{7. 16}$$

and the non-isotropic ball

$$\mathbf{B}_{\lambda,\delta} ~=~ \Big\{(x,y)\in\mathbb{R}^{\mathbf{N}_1}\times\mathbb{R}^{\mathbf{N}_2}:~\lambda(x,y)\leq\delta\Big\}. \tag{7. 17}$$

Let $(u,v)\in\mathbf{B}_{\lambda,\delta}$ and $(x,y)\in{}^c\mathbf{B}_{\lambda,r\delta}$ for some $r>1$. By adjusting its value, we can have

$$\lambda(x,y) ~>~ r\delta, \qquad \lambda(x-u,y-v) ~>~ \delta. \tag{7. 18}$$

The differential inequality in (7. 7) implies

$$\begin{aligned}
&\Big|\Omega(x-u,y-v)-\Omega(x,y)\Big|_H\\
&\leq~ \mathfrak{B}~\frac{|u|}{\lambda(x,y)}\left(\frac{1}{\lambda(x,y)}\right)^{\mathbf{N}_1+\mathbf{N}_2/\rho}+\mathfrak{B}~\frac{|v|}{\lambda(x,y)^{1/\rho}}\left(\frac{1}{\lambda(x,y)}\right)^{\mathbf{N}_1+\mathbf{N}_2/\rho}\\
&\leq~ \mathfrak{B}~\frac{\lambda(u,v)}{\lambda(x,y)}\left(\frac{1}{\lambda(x,y)}\right)^{\mathbf{N}_1+\mathbf{N}_2/\rho}.
\end{aligned}\tag{7. 19}$$

From (7. 18) and (7. 19), we have

$$\begin{aligned}
&\sum_{k=0}^\infty\iint_{\mathbf{B}_{\lambda,2^{k+1}r\delta}\setminus\mathbf{B}_{\lambda,2^kr\delta}}\Big|\Omega(x-u,y-v)-\Omega(x,y)\Big|_H\,dxdy\\
&\leq~\mathfrak{B}\sum_{k=0}^\infty\iint_{\mathbf{B}_{\lambda,2^{k+1}r\delta}\setminus\mathbf{B}_{\lambda,2^kr\delta}}\frac{\lambda(u,v)}{\lambda(x,y)}\left(\frac{1}{\lambda(x,y)}\right)^{\mathbf{N}_1+\mathbf{N}_2/\rho}dxdy\\
&\leq~\mathfrak{B}\,2^{\mathbf{N}_1+\mathbf{N}_2/\rho}\sum_{k=0}^\infty 2^{-k}, \qquad (u,v)\in\mathbf{B}_{\lambda,\delta}.
\end{aligned}\tag{7. 20}$$

From (7. 20), for every $(u,v)\in \mathbf{B}_{\lambda,\delta}$, we conclude

$$\iint_{{}^c\mathbf{B}_{\lambda,r\delta}} \Big|\Omega(x-u,y-v)-\Omega(x,y)\Big|_H\, dxdy \ \leq\ \mathfrak{B}. \tag{7. 21}$$

This is a well known condition for $f*\Omega$ satisfying the weak type $(1,1)$-estimate with the Hilbert space valued norm given in (7. 5)-(7. 6).

On the other hand, by applying Plancherel theorem, we have

$$\begin{aligned}
\Big\|\mathfrak{S}_\Phi f\Big\|^2_{\mathbf{L}^2(\mathbb{R}^{\mathbf{N}_1}\times\mathbb{R}^{\mathbf{N}_2})} \ &=\ \iint_{\mathbb{R}^{\mathbf{N}_1}\times\mathbb{R}^{\mathbf{N}_2}} \left\{\int_0^\infty |\widehat{f}(\xi,\eta)|^2 \left|\widehat{\Phi}\left(\delta\xi,\delta^{\frac{1}{\rho}}\eta\right)\right|^2 \frac{d\delta}{\delta}\right\} d\xi d\eta \\
&\leq\ \left\{\sup_{(\xi,\eta)} \int_0^\infty \phi^2\left(\delta\xi,\delta^{\frac{1}{\rho}}\eta\right)\frac{d\delta}{\delta}\right\} \iint_{\mathbb{R}^{\mathbf{N}_1}\times\mathbb{R}^{\mathbf{N}_2}} |\widehat{f}(\xi,\eta)|^2 d\xi d\eta \qquad (7.\ 22)\\
&\leq\ \mathfrak{B}\ \Big\|f\Big\|^2_{\mathbf{L}^2(\mathbb{R}^{\mathbf{N}_1}\times\mathbb{R}^{\mathbf{N}_2})}.
\end{aligned}$$

From the weak type $(1,1)$-estimate and (7. 22), we conclude the $\mathbf{L}^p$-result in (7. 4) by using Marcinkewicz interpolation theorem and a standard argument of duality.

Consider $\widehat{\Psi}_\delta(\xi,\eta)=\phi\left(\delta^{\frac{1}{\rho}}\xi,\delta\eta\right)$. Our estimates proving **Lemma 7.1** remain valid for $\Omega \doteq \Psi_\delta$, with x and y switched in roles. Denote $F=f*\Phi_\delta$. By applying the $\mathbf{L}^p$-regularity theorem for space valued functions, stated as Theorem 3 in Chapter I of Stein [7], we have

$$\begin{aligned}
\iint_{\mathbb{R}^{\mathbf{N}_1}\times\mathbb{R}^{\mathbf{N}_2}} \left(\mathfrak{S}_{\Phi*\Psi} f\right)^p(x,y)dxdy \ &=\ \iint_{\mathbb{R}^{\mathbf{N}_1}\times\mathbb{R}^{\mathbf{N}_2}} \left\{\int_0^\infty \Big|F*\Psi_\delta(x,y)\Big|^2_H \frac{d\delta}{\delta}\right\}^{\frac{p}{2}} dxdy \\
&\leq\ \mathfrak{B}_p\ \iint_{\mathbb{R}^{\mathbf{N}_1}\times\mathbb{R}^{\mathbf{N}_2}} \Big|F(x,y)\Big|^p_H\, dxdy \\
&=\ \mathfrak{B}_p\ \iint_{\mathbb{R}^{\mathbf{N}_1}\times\mathbb{R}^{\mathbf{N}_2}} \left\{\int_0^\infty \Big|f*\Phi_\delta(x,y)\Big|^2 \frac{d\delta}{\delta}\right\}^{\frac{p}{2}} dxdy \qquad (7.\ 23)\\
&\leq\ \mathfrak{B}_p\ \iint_{\mathbb{R}^{\mathbf{N}_1}\times\mathbb{R}^{\mathbf{N}_2}} \Big|f(x,y)\Big|^p\, dxdy, \qquad 1<p<\infty.
\end{aligned}$$

Lastly, it is clear that the above iteration argument works for any number of parameter. Replace $\mathbf{N}_1,\mathbf{N}_2$ with $\mathbf{N}_i,\mathbf{N}-\mathbf{N}_i$ for every $i=1,2,\ldots,n$ in (7. 1)-(7. 22). By carrying out an n-parameter analogue of (7. 23), we conclude the $\mathbf{L}^p$-boundedness of the corresponding square function. This is equivalent to our desired Littlewood-Paley inequality in (6. 6).

References

[1] A. Nagel, F. Ricci, E. M. Stein and S. Wainger *Algebras of Singular Integral Operators with Kernels Controlled by Multiple Norms*, Memoirs of the American Mathematical Society, Vol 256: No.1230, 2018.

[2] A. Nagel, F. Ricci and E. M. Stein, *Singular Integrals with Flag Kernels and Analysis on Quadratic CR Manifolds*, Journal of Functional Analysis **181**: 29-118, 2001.

[3] D. Müller, F. Ricci, E. M. Stein, *Marcinkiewicz Multipliers and Multi-parameter structures on Heisenberg (-type) group, I*, Inventiones Mathematicae **119**: 199-233, 1995.

[4] F. Ricci and E. M. Stein, *Multiparameter Singular Integrals and Maximal Functions*, Annales de L' institute Fourier, **42**, 637-670, 1992.

[5] C. Fefferman and E. M. Stein, Some Maximal Inequalities, American Journal of Mathematics **93**: 107-115, 1971.

[6] R. Fefferman and E. M. Stein, *Singular Integrals on Product Spaces*, Advances in Mathematics **45**:117-143, 1982.

[7] E. M. Stein, *Harmonic Analysis: Real-Variable Methods, Orthogonality and Oscillatory Integrals*, Princeton University Press, 1993.

[8] A. Nagel and E. M. Stein, *On the Product Theory of Singular Integrals*, Revista Matemática Iberoamericana **20**: 531-561, 2004.

[9] R. Fefferman, *Harmonic Analysis on Product Spaces*, Annals of Mathematics **126**: 109-130, 1987.

[10] R. Fefferman, *Bounded Mean oscillation on the polydisc*, Annals of Mathematics **110**: no.2, 395-406, 1979.

[11] A. Córdoba and R. Fefferman, *A geometric Proof of the Strong Maximal Theorem*, Annals of Mathematics **102**: 95-100, 1975.

[12] S. Y. A. Chang and R. Fefferman, *The Colderón-Zygmund Decomposition on Product Domains*, American Journal of Mathematics **104**: 455-468, 1982.

[13] J. L. Journé, *Calderón-Zygmund Operators on Product Spaces*, Revista Matemática Iberoamericana **1**: no.3, 55-91, 1985.

[14] J. Pipher, *Journé's Covering Lemma and Its Extension to Higher Dimensions*, Duke Mathematics Journal **53**: no.3, 683-690, 1986.

[15] L. Boutet de Monvel, *Hypoelliptic Operators with Double Characteristics and Related Pseudo Differential Operators*, Communication in Pure and Applied Mathematics **27**: 585-639, 1974.

[16] R. Beals and C. Fefferman, *Spatially Inhomogeneous Pseudo Differential Operators*, Communication in Pure and Applied Mathematics **27**: 1-24, 1974.

[17] A. Carbery and A. Seeger, *Conditionally Convergent Series of Linear Operators on* $\mathbf{L}^p$*-Spaces and* $\mathbf{L}^p$*-Estimates for Pseudodifferential Operators*, Proceeding of London Mathematical Society, **57**(3), 481-510, 1988.

[18] A. Carbery and A. Seeger, $\mathbf{H}^p$ *and* $\mathbf{L}^p$ *Variants of Multiparameter Calderón-Zygmund Theory*, Transaction of the American Mathematical Society **334**: 719-747, 1992.

School of Science, Westlake University, Hangzhou, 310030, China
email: wangzipeng@westlake.edu.cn